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Foreword

This second edition European Telecommunication Standard (ETS) has been produced by the Transmission and Multiplexing (TM) Technical Committee of the European Telecommunications Standards Institute (ETSI).

This second edition ETS describes the information model for Network Elements (NEs) which use the Synchronous Digital Hierarchy (SDH) multiplexing structure.

Transposition dates	
Date of adoption	18 October 1996
Date of latest announcement of this ETS (doa):	31 May 1997
Date of latest publication of new National Standard or endorsement of this ETS (dop/e):	30 November 1997
Date of withdrawal of any conflicting National Standard (dow):	30 November 1997

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1 Scope

This second edition ETS defines the information model to be used at the interface between Network Elements (NEs) and management systems, for the management of Synchronous Digital Hierarchy (SDH) NEs.

This ETS defines the information model for SDH NEs.

This ETS does not define:

- the protocol stack to be used for message communication;
- the network level management processes;
- the application contexts;
- the conformance requirements to be met by an implementation of this information model;
- information models for other systems or equipment.

The information model defined here (and the corresponding message set) is concerned with the management of NEs, the equipment by which they are implemented and the functions contained within them. More precisely, it applies to an equipment domain visible at the element manager to NE interface and is only concerned with information available within that domain. Information proper to the domain of a network level management process is not included within this model.

2 Normative references

This ETS incorporates, by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this ETS only when incorporated in it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

[1] ITU-T Recommendation G.709 (1993): "Synchronous multiplexing structure".

[2] ITU-T Recommendation G.774 (1992): "Synchronous digital hierarchy (SDH) management information model for the network element view".

[3] ITU-T Recommendation G.774.03 (1994): "Synchronous digital hierarchy (SDH) management of multiplex-section protection for the network element view".

[4] ITU-T Recommendation G.783 (1993): "Characteristics of synchronous digital hierarchy (SDH) equipment functional blocks".

[5] ITU-T Recommendation M.3100 (1995): "Generic network information model".

[6] ITU-T Recommendation X.721 (1992): "Information technology - Open Systems Interconnection - Structure of management information: Definition of management information".

[7] ITU-T Recommendation G.702 (1988): "Digital hierarchy bit rates".

[8] ETS 300 371 (1994): "Transmission and Multiplexing (TM); Plesiochronous Digital Hierarchy (PDH) information model for the Network Element (NE) view".

3 Abbreviations

For the purposes of this ETS, the following abbreviations apply:

AIS	Alarm Indication Signal
AP	Access Point
ATM	Asynchronous Transfer Mode
AU	Administrative Unit
AUG	Administrative Unit Group
CCITT	Comité Consultatif International Télégraphique et Téléphonique
CMIP	Common Management Information Protocol
CMIS	Common Management Information Service
CP	Connection Point
CTP	Connection Termination Point
GTP	Group Termination Point
HPA	Higher Order Path Adaptation
HPC	Higher Order Path Connection
HPT	Higher Order Path Termination
IA	Indirect Adaptor
IOS	Intra-Office Section
ISO	International Organization for Standardization
ITU-T	International Telecommunications Union - Telecommunications sector
LOF	Loss Of Frame
LPA	Lower Order Path Adaptation
LPC	Lower Order Path Connection
LPT	Lower Order Path Termination
MS	Multiplexer Section
MSA	Multiplexer Section Adaptation
MST	Multiplexer Section Termination
MSTTP	Multiplexer Section Trail Termination Point
NE	Network Element
OS	Operation System
OSI	Open Systems Interconnection
PDH	Plesiochronous Digital Hierarchy
Pkg	Packages
POH	Path OverHead
PPI	Plesiochronous Physical Interface
RDN	Relative Distinguished Name
RS	Regenerator Section
RST	Regenerator Section Termination
RSTTP	Regenerator Section Trail Termination Point
SDH	Synchronous Digital Hierarchy
SDHNE	Synchronous Digital Hierarchy Network Element
Snk	Sink
Src	Source
SPI	Synchronous Physical Interface
STM-N	Synchronous Transport Module-N
TMN	Telecommunication Management Network
TP	Termination Point
TTP	Trail Termination Point
TU	Tributary Unit
TUG	Tributary Unit Group
VC-n	Virtual Container n

4 Registration supporting Abstract Syntax Notation one (ASN.1)

```
PrETS5 {ccitt(0) identified-organization(4) etsi(0) ets304(304) informationModel(0) asn1Module(2)
prETS5(0)}
DEFINITIONS IMPLICIT TAGS ::= BEGIN
-- EXPORT Everything
prETS300304 OBJECT IDENTIFIER ::= {ccitt(0) identified-organization(4) etsi(0) ets304(304)
informationModel(0)}
etsObjectClass OBJECT IDENTIFIER ::= {prETS300304 managedObjectClass(3)}
etsPackage OBJECT IDENTIFIER ::= {prETS300304 package(4)}
etsParameter OBJECT IDENTIFIER ::= {prETS300304 parameter(5)}
etsNameBinding OBJECT IDENTIFIER ::= {prETS300304 nameBinding(6)}
etsAttribute OBJECT IDENTIFIER ::= {prETS300304 attribute(7)}
etsAction OBJECT IDENTIFIER ::= {prETS300304 action(9)}
etsNotification OBJECT IDENTIFIER ::= {prETS300304 notification(10)}
END
```

5 Generic objects fragment

In this fragment, a working sub-set of standard and mature object classes have been adopted, mainly from the ITU-T X.700 series of Recommendations.

5.1 Generic objects - object classes

In this context the IMPORTS clause specifies the object classes which can be instantiated in the scope of this ETS. The IMPORTS clause does not include uninstantiated superclasses.

```
BEGIN
IMPORTS
alarmRecord,
attributeValueChangeRecord,
eventForwardingDiscriminator,
log,
objectCreationRecord
objectDeletionRecord
stateChangeRecord,
FROM {joint-iso-ccitt ms(9) smi(3) part2(2) managedObjectClass(3) }

alarmSeverityAssignmentProfile
FROM {ccitt(0) recommendation(0) m(13) m3100(3100) informationModel(0) managedObjectClass(3) }
;
END
```

5.2 Generic objects - packages, attributes, ASN.1, name-bindings

All packages, attributes, ASN.1 and name-bindings associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

6 SDH Termination Point (TP) fragment

6.1 SDH TP - object classes

In this context, the IMPORTS clause specifies the object classes which can be instantiated in the scope of this ETS. The IMPORTS clause does not include uninstantiated superclasses.

```
BEGIN
IMPORTS
au4CTPBidirectional,
au4CTPSink,
au4CTPSource,
augBidirectional,
augSink,
augSource,
electricalSPITTPBidirectional,
electricalSPITTPSink,
electricalSPITTPSource,
msCTPBidirectional,
msCTPSink,
msCTPSource,
msDatacomCTPBidirectional,
msDatacomCTPSink,
msDatacomCTPSource,
msOrderwireCTPBidirectional,
```

```
msOrderwireCTPSink,
msOrderwireCTPSource,
msTTPBidirectional,
msTTPSink,
msTTPSource,
opticalSPITTPBidirectional,
opticalSPITTPSink,
opticalSPITTPSource,
rsCTPBidirectional,
rsCTPSink,
rsCTPSource,
rsDatacomCTPBidirectional,
rsDatacomCTPSink,
rsDatacomCTPSource,
rsOrderwireCTPBidirectional,
rsOrderwireCTPSink,
rsOrderwireCTPSource,
rsTTPBidirectional,
rsTTPSink,
rsTTPSource,
rsUserChannelCTPBidirectional,
rsUserChannelCTPSink,
rsUserChannelCTPSource,
tu11CTPBidirectional,
tu11CTPSink,
tu11CTPSource,
tu12CTPBidirectional,
tu12CTPSink,
tu12CTPSource,
tu2CTPBidirectional,
tu2CTPSink,
tu2CTPSource,
tu3CTPBidirectional,
tu3CTPSink,
tu3CTPSource,
tug2Bidirectional,
tug2Sink,
tug2Source,
tug3Bidirectional,
tug3Sink,
tug3Source,
vc11TTPBidirectional,
vc11TTPSink,
vc11TTPSource,
vc12TTPBidirectional,
vc12TTPSink,
vc12TTPSource,
vc2TTPBidirectional,
vc2TTPSink,
vc2TTPSource,
vc3TTPBidirectional,
vc3TTPSink,
vc3TTPSource,
vc4TTPBidirectional,
vc4TTPSink,
vc4TTPSource,
vcnUserChannelCTPBidirectional,
vcnUserChannelCTPSink,
vcnUserChannelCTPSource
FROM {ccitt(0) recommendation(0) g(7) g774(774) informationModel(0) managedObjectClass(3) }
;
END
```

6.2 SDH TP - packages

All packages associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

6.3 SDH TP - attributes

All attributes associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

```
supportedByObjectList
    The value of the supportedByObjectList attribute points to the equipment and software
    objects which implement the TPs.
```

6.4 SDH TP - name bindings

```
BEGIN
IMPORTS
au4CTPBidirectional-augBidirectional,
au4CTPSink-augBidirectional,
au4CTPSink-augSink,
au4CTPSource-augBidirectional,
au4CTPSource-augSource,
augBidirectional-msTTPBidirectional,
augSink-msTTPSink,
augSource-msTTPSource,
electricalSPITTPBidirectional-sdhNE,
electricalSPITTPSink-sdhNE,
electricalSPITTPSource-sdhNE,
msCTPBidirectional-rsTTPBidirectional,
msCTPSink-rsTTPBidirectional,
msCTPSink-rsTTPSink,
msCTPSource-rsTTPBidirectional,
msCTPSource-rsTTPSource,
msDatacomCTPBidirectional-msTTPBidirectional,
msDatacomCTPSink-msTTPBidirectional,
msDatacomCTPSink-msTTPSink,
msDatacomCTPSource-msTTPBidirectional,
msDatacomCTPSource-msTTPSource,
msOrderwireCTPBidirectional-msTTPBidirectional,
msOrderwireCTPSink-msTTPBidirectional,
msOrderwireCTPSink-msTTPSink,
msOrderwireCTPSource-msTTPBidirectional,
msOrderwireCTPSource-msTTPSource,
msTTPBidirectional-sdhNE,
msTTPSink-sdhNE,
msTTPSource-sdhNE,
opticalSPITTPBidirectional-sdhNE,
opticalSPITTPSink-sdhNE,
opticalSPITTPSource-sdhNE,
rsCTPBidirectional-electricalSPITTPBidirectional,
rsCTPBidirectional-opticalSPITTPBidirectional,
rsCTPSink-electricalSPITTPBidirectional,
rsCTPSink-electricalSPITTPSink,
rsCTPSink-opticalSPITTPBidirectional,
rsCTPSink-opticalSPITTPSink,
rsCTPSource-electricalSPITTPBidirectional,
rsCTPSource-electricalSPITTPSource,
rsCTPSource-opticalSPITTPBidirectional,
rsCTPSource-opticalSPITTPSource,
rsDatacomCTPBidirectional-rsTTPBidirectional,
rsDatacomCTPSink-rsTTPBidirectional,
rsDatacomCTPSink-rsTTPSink,
rsDatacomCTPSource-rsTTPBidirectional,
rsDatacomCTPSource-rsTTPSource,
rsOrderwireCTPBidirectional-rsTTPBidirectional,
rsOrderwireCTPSink-rsTTPBidirectional,
rsOrderwireCTPSink-rsTTPSink,
rsOrderwireCTPSource-rsTTPBidirectional,
rsOrderwireCTPSource-rsTTPSource,
rsTTPBidirectional-sdhNE,
rsTTPSink-sdhNE,
rsTTPSource-sdhNE,
rsUserChannelCTPBidirectional-rsTTPBidirectional,
rsUserChannelCTPSink-rsTTPBidirectional,
rsUserChannelCTPSink-rsTTPSink,
rsUserChannelCTPSource-rsTTPBidirectional,
rsUserChannelCTPSource-rsTTPSource,
tu11CTPBidirectional-tug2Bidirectional,
tu11CTPSink-tug2Bidirectional,
tu11CTPSink-tug2Sink,
tu11CTPSource-tug2Bidirectional,
tu11CTPSource-tug2Source,
tu12CTPBidirectional-tug2Bidirectional,
tu12CTPSink-tug2Bidirectional,
tu12CTPSink-tug2Sink,
tu12CTPSource-tug2Bidirectional,
tu12CTPSource-tug2Source,
tu2CTPBidirectional-tug2Bidirectional,
tu2CTPSink-tug2Bidirectional,
tu2CTPSink-tug2Sink,
tu2CTPSource-tug2Bidirectional,
tu2CTPSource-tug2Source,
tu3CTPBidirectional-tug3Bidirectional,
tu3CTPSink-tug3Bidirectional,
tu3CTPSink-tug3Sink,
tu3CTPSource-tug3Bidirectional,
tu3CTPSource-tug3Source,
```

```
tug2Bidirectional-tug3Bidirectional,
tug2Sink-tug3Sink,
tug2Source-tug3Source,
tug3Bidirectional-vc4TTPBidirectional,
tug3Sink-vc4TTPSink,
tug3Source-vc4TTPSource,
vc11TTPBidirectional-sdhNE,
vc11TTPSink-sdhNE,
vc11TTPSource-sdhNE,
vc12TTPBidirectional-sdhNE,
vc12TTPSink-sdhNE,
vc12TTPSource-sdhNE,
vc2TTPBidirectional-sdhNE,
vc2TTPSink-sdhNE,
vc2TTPSource-sdhNE,
vc3TTPBidirectional-sdhNE,
vc3TTPSink-sdhNE,
vc3TTPSource-sdhNE,
vc4TTPBidirectional-sdhNE,
vc4TTPSink-sdhNE,
vc4TTPSource-sdhNE,
vcnUserChannelCTPBidirectional-vc3TTPBidirectional,
vcnUserChannelCTPBidirectional-vc4TTPBidirectional,
vcnUserChannelCTPSink-vc3TTPBidirectional,
vcnUserChannelCTPSink-vc3TTPSink,
vcnUserChannelCTPSink-vc4TTPBidirectional,
vcnUserChannelCTPSink-vc4TTPSink,
vcnUserChannelCTPSource-vc3TTPBidirectional,
vcnUserChannelCTPSource-vc3TTPSource,
vcnUserChannelCTPSource-vc4TTPBidirectional,
vcnUserChannelCTPSource-vc4TTPSource
FROM {ccitt(0) recommendation(0) g(7) g774(774) informationModel(0) nameBinding(6) }
;
END
```

6.5 SDH TP - subordination rules

```
BEGIN
IMPORTS
augSinkSubordination,
augSourceSubordination,
augBidirectionalSubordination,
electricalSPITTPSinkSubordination,
electricalSPITTPSourceSubordination,
electricalSPITTPBidirectionalSubordination,
opticalSPITTPSinkSubordination,
opticalSPITTPSourceSubordination,
opticalSPITTPBidirectionalSubordination,
msTTPSinkSubordination,
msTTPSourceSubordination,
msTTPBidirectionalSubordination,
rsTTPSinkSubordination,
rsTTPSourceSubordination,
rsTTPBidirectionalSubordination,
sdhNESubordination,
tug2SinkSubordination,
tug2SourceSubordination,
tug2BidirectionalSubordination,
tug3SinkSubordination,
tug3SourceSubordination,
tug3BidirectionalSubordination,
vc3TTPSinkSubordination,
vc3TTPSourceSubordination,
vc3TTPBidirectionalSubordination,
vc4TTPSinkSubordination,
vc4TTPSourceSubordination,
vc4TTPBidirectionalSubordination,

FROM {ccitt(0) recommendation(0) g(7) g774(774)}
;
END
```

6.6 SDH TP - constraints

```
BEGIN
IMPORTS
downstreamConnectivityPointer-au4CTPSink,
upstreamConnectivityPointer-au4CTPSource,
downstreamConnectivityPointer-msCTPSink,
upstreamConnectivityPointer-msCTPSource,
upstreamConnectivityPointer-msTTPSink,
downstreamConnectivityPointer-msTTPSource,
downstreamConnectivityPointer-rsCTPSink,
upstreamConnectivityPointer-rsCTPSource,
upstreamConnectivityPointer-rsTTPSink,
downstreamConnectivityPointer-rsTTPSource,
downstreamConnectivityPointer-tu11CTPSink,
upstreamConnectivityPointer-tu11CTPSource,
downstreamConnectivityPointer-tu12CTPSink,
upstreamConnectivityPointer-tu12CTPSource,
downstreamConnectivityPointer-tu2CTPSink,
upstreamConnectivityPointer-tu2CTPSource,
downstreamConnectivityPointer-tu3CTPSink,
upstreamConnectivityPointer-tu3CTPSource,
upstreamConnectivityPointer-vc11TTPSink,
downstreamConnectivityPointer-vc11TTPSource,
upstreamConnectivityPointer-vc12TTPSink,
downstreamConnectivityPointer-vc12TTPSource,
upstreamConnectivityPointer-vc2TTPSink,
downstreamConnectivityPointer-vc2TTPSource,
upstreamConnectivityPointer-vc3TTPSink,
downstreamConnectivityPointer-vc3TTPSource,
FROM {ccitt(0) recommendation(0) g(7) g774(774)}
;
END

ets_upstreamConnectivityPointer-vc4TTPSink CONSTRAINT RULE
  OBJECT CLASS
     vc4TTPSink AND SUBCLASSES;
  IS RELATED TO
     vc4TTPSource, vc4TTPBidirectional,
     au4CTPSink, au4CTPBidirectional;
  USING ATTRIBUTE
     "Recommendation M.3100":upstreamConnectivityPointer;
  ACCORDING TO RULE
     SET SIZE (1) OF CHOICE{
       vc4TTPSource, vc4TTPBidirectional,
       au4CTPSink, au4CTPBidirectional};
;

ets_downstreamConnectivityPointer-vc4TTPSource CONSTRAINT RULE
  OBJECT CLASS
     vc4TTPSource AND SUBCLASSES;
  IS RELATED TO
     vc4TTPSink, vc4TTPBidirectional,
     au4CTPSource, au4CTPBidirectional;
  USING ATTRIBUTE
     "Recommendation M.3100":downstreamConnectivityPointer;
  CASE{
    single ACCORDING TO RULE
     SET SIZE (1) OF CHOICE{
     vc4TTPSink, vc4TTPBidirectional,
     au4CTPSource, au4CTPBidirectional};
;   broadcast ACCORDING TO RULE
     SET SIZE (1..N) OF CHOICE{
     vc4TTPSink, vc4TTPBidirectional,
     au4CTPSource, au4CTPBidirectional};
}
```

7 Plesiochronous Digital Hierarchy (PDH) fragment

This clause provides managed objects required to model PDH interfaces on SDH equipment.

7.1 Object classes definitions

```
ITU-T Recommendation G.702 2 Mbit/s connection termination point

p12CTPSink  MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointSink;
   CHARACTERIZED BY
       "Recommendation M.3100":createDeleteNotificationsPackage,
       "Recommendation M.3100":operationalStatePackage,
       "Recommendation M.3100":stateChangeNotificationPackage,
       p12CTPSinkPkg  PACKAGE
         BEHAVIOUR
            p12CTPSinkBehaviourPkg BEHAVIOUR
               DEFINED AS
                  *This managed object class terminates a G.702 2 Mbit/s connection and includes
the lower order path adaptation function (LPA).*
;;
         ATTRIBUTES
            p12CTPId                                    GET;
;;
   REGISTERED AS { etsObjectClass 13 };


p12CTPSource MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointSource;
   CHARACTERIZED BY
       "Recommendation M.3100":createDeleteNotificationsPackage,
       p12CTPSourcePkg  PACKAGE
        BEHAVIOUR
            p12CTPSourceBehaviourPkg BEHAVIOUR
               DEFINED AS
                  *This object class originates a G.702 2 Mbit/s connection*
;;
         ATTRIBUTES
            p12CTPId                                    GET;
;;
   REGISTERED AS { etsObjectClass 2 };


p12CTPBidirectional MANAGED OBJECT CLASS
DERIVED FROM "Recommendation M.3100":connectionTerminationPointBidirectional,
                        p12CTPSink,
                        p12CTPSource;
  REGISTERED AS { etsObjectClass 3 };


G702 34 Mbit/s connection termination point

p31CTPSink  MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointSink;
   CHARACTERIZED BY
       "Recommendation M.3100":createDeleteNotificationsPackage,
       "Recommendation M.3100":operationalStatePackage,
       "Recommendation M.3100":stateChangeNotificationPackage,
       p31CTPSinkPkg  PACKAGE
         BEHAVIOUR
            p31CTPSinkBehaviourPkg BEHAVIOUR
               DEFINED AS
                  *This managed object class terminates a G.702 34 Mbit/s connection and includes
the lower order path adaptation function (LPA).*
;;
         ATTRIBUTES
            p31CTPId                                    GET;
;;
   REGISTERED AS { etsObjectClass 4 };


p31CTPSource MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointSource;
   CHARACTERIZED BY
       "Recommendation M.3100":createDeleteNotificationsPackage,
       p31CTPSourcePkg  PACKAGE
        BEHAVIOUR
            p31CTPSourceBehaviourPkg BEHAVIOUR
               DEFINED AS
                  *This object class originates a G.702 34Mbit/s connection.*
;;
         ATTRIBUTES
```

```
           p31CTPId                                    GET;
;;
   REGISTERED AS { etsObjectClass 5 };


p31CTPBidirectional MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointBidirectional,
                        p31CTPSink,
                        p31CTPSource;
  REGISTERED AS { etsObjectClass 6 };


G702 140 Mbit/s connection termination point

p4CTPSink  MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointSink;
   CHARACTERIZED BY
       "Recommendation M.3100":createDeleteNotificationsPackage,
       "Recommendation M.3100":operationalStatePackage,
       "Recommendation M.3100":stateChangeNotificationPackage,
       p4CTPSinkPkg  PACKAGE
         BEHAVIOUR
            p4CTPSinkBehaviourPkg BEHAVIOUR
               DEFINED AS
                  *This managed object class terminates a G.702 140 Mbit/s connection and
includes the lower order path adaptation function (LPA).*
;;
         ATTRIBUTES
            p4CTPId                                  GET;
;;
   REGISTERED AS { etsObjectClass 7 };


p4CTPSource MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointSource;
   CHARACTERIZED BY
       "Recommendation M.3100":createDeleteNotificationsPackage,
       p4CTPSourcePkg  PACKAGE
        BEHAVIOUR
            p4CTPSourceBehaviourPkg BEHAVIOUR
               DEFINED AS
                  *This object class originates a G.702 140 Mbit/s connection.*
;;
         ATTRIBUTES
            p4CTPId                                  GET;
;;
   REGISTERED AS { etsObjectClass 8 };


p4CTPBidirectional MANAGED OBJECT CLASS
  DERIVED FROM "Recommendation M.3100":connectionTerminationPointBidirectional,
                        p4CTPSink,
                        p4CTPSource;
  REGISTERED AS { etsObjectClass 9 };
```

7.2 Attributes definitions

```
supportedByObjectList
The value of the supportedByObjectList attribute points to the equipment and software objects
which implement the TPs.

G.702 2Mbit/s Connection Termination Point Identification

p12CTPId ATTRIBUTE
   WITH ATTRIBUTE SYNTAX       PrETS8.NameType ;
   MATCHES FOR                 EQUALITY;
   BEHAVIOUR
    p12CTPIdBehaviour BEHAVIOUR
    DEFINED AS
    This attribute is used as an RDN for naming instances of the p12CTP object classes.
;;
REGISTERED AS { etsAttribute 1 };

G.702 34Mbit/s Connection Termination Point Identification

p31CTPId ATTRIBUTE
   WITH ATTRIBUTE SYNTAX       PrETS8.NameType ;
   MATCHES FOR                 EQUALITY;
   BEHAVIOUR
    p31CTPIdBehaviour BEHAVIOUR
    DEFINED AS
    This attribute is used as an RDN for naming instances of the p31CTP object classes.
;;
REGISTERED AS { etsAttribute 2 };


G.702 140Mbit/s Connection Termination Point Identification

p4CTPId ATTRIBUTE
   WITH ATTRIBUTE SYNTAX       PrETS8.NameType ;
   MATCHES FOR                 EQUALITY;
   BEHAVIOUR
      p4CTPIdBehaviour BEHAVIOUR
      DEFINED AS
         This attribute is used as an RDN for naming instances of the p4CTP object classes.
;;
REGISTERED AS { etsAttribute 3 };
```

7.3 Name bindings definitions

```
p12CTPSource-G774vc12TTPSource NAME BINDING
   SUBORDINATE OBJECT CLASS p12CTPSource AND SUBCLASSES;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":vc12TTPSource AND SUBCLASSES;
   WITH ATTRIBUTE           p12CTPId;
   BEHAVIOUR
      p12CTPSource-vc12TTPSource  BEHAVIOUR
    DEFINED AS
    The subordinate managed object may be automatically instantiated when the superior managed
object is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 20 };

p12CTPSink-G774vc12TTPSink NAME BINDING
   SUBORDINATE OBJECT CLASS p12CTPSink AND SUBCLASSES;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":vc12TTPSink AND SUBCLASSES;
   WITH ATTRIBUTE           p12CTPId;
   BEHAVIOUR
      p12CTPSink-vc12TTPSink  BEHAVIOUR
    DEFINED AS
    The subordinate managed object may be automatically instantiated when the superior managed
object is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 21 };


p31CTPSource-G774vc3TTPSource NAME BINDING
   SUBORDINATE OBJECT CLASS p31CTPSource AND SUBCLASSES;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":vc3TTPSource AND SUBCLASSES;
   WITH ATTRIBUTE           p31CTPId;
   BEHAVIOUR
    p31CTPSource-vc3TTPSource BEHAVIOUR
    DEFINED AS
    The subordinate managed object may be automatically instantiated when the superior managed
object is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 22 };

p31CTPSink-G774vc3TTPSink NAME BINDING
   SUBORDINATE OBJECT CLASS p31CTPSink AND SUBCLASSES;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":vc3TTPSink AND SUBCLASSES;
   WITH ATTRIBUTE           p31CTPId;
   BEHAVIOUR
    p31CTPSink-vc3TTPSink  BEHAVIOUR
    DEFINED AS
    The subordinate managed object may be automatically instantiated when the superior managed
object is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 23 };


p4CTPSource-G774vc4TTPSource NAME BINDING
   SUBORDINATE OBJECT CLASS p4CTPSource AND SUBCLASSES;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":vc4TTPSource AND SUBCLASSES;
   WITH ATTRIBUTE           p4CTPId;
   BEHAVIOUR
      p4CTPSource-vc4TTPSource  BEHAVIOUR
    DEFINED AS
    The subordinate managed object may be automatically instantiated when the superior managed
object is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 24 };

p4CTPSink-G774vc4TTPSink NAME BINDING
   SUBORDINATE OBJECT CLASS p4CTPSink AND SUBCLASSES;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":vc4TTPSink AND SUBCLASSES;
   WITH ATTRIBUTE           p4CTPId;
   BEHAVIOUR
    p4CTPSink-vc4TTPSink  BEHAVIOUR
    DEFINED AS
    The subordinate managed object may be automatically instantiated when the superior managed
object is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 25 };
```

7.4 ASN.1 definitions

```
PrETS8{ccitt(0) identified-organization(4) etsi(0) ets304(304) informationModel(0) asn1Module(2)
prETS8(1)}
DEFINITIONS IMPLICIT TAGS ::=
BEGIN
-- EXPORTS everything

IMPORTS

NameType FROM ASN1DefinedTypesModule {ccitt(0) recommendation(0) m(13) m3100(3100)
informationModel(0) asn1Module(2) asn1DefinedTypesModule(0)}

END -- end of ASN1DefinedTypesModule
```

8 Cross-connection fragment

8.1 Cross-connection - object classes

In this context the IMPORTS clause specifies the object classes which can be instantiated in the scope of this ETS. The IMPORTS clause does not include uninstantiated superclasses.

```
BEGIN
IMPORTS
crossConnection,
fabric,
gtp,
mpCrossConnection,
tpPool
FROM {ccitt(0) recommendation(0) m(13) m3100(3100) informationModel(0) managedObjectClass(3) }
;
END
```

8.2 Cross-connection - packages

All packages associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

8.3 Cross-connection - attributes

All attributes associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

```
supportedByObjectList
    The value of the supportedByObjectList attribute points to the equipment and software
    objects which implement the TPs.
```

8.4 Cross-connection - name bindings

```
BEGIN
IMPORTS
fabric-managedElement,
gtp-fabric,
mpCrossConnection-fabric,
tpPool-fabric
FROM {ccitt(0) recommendation(0) m(13) m3100(3100) informationModel(0) nameBinding(6) }
;
END

etsCrossConnection-fabric NAME BINDING
SUBORDINATE OBJECT CLASS "Recommendation M.3100" : crossConnection
  AND SUBCLASSES;
NAMED BY
SUPERIOR OBJECT CLASS    "Recommendation M.3100":fabric
  AND SUBCLASSES;
WITH ATTRIBUTE           "Recommendation M.3100":crossConnectionId;
BEHAVIOUR
  etsCrossConnection-fabricBehaviour BEHAVIOUR
     DEFINED AS
*The value of the fromTermination attribute in the crossConnection object shall not be NULL. When
an instance of crossConnection is deleted, the following attributes will be affected. The
crossConnectionObjectPointer attributes in the termination points or in the gtp objects that were
pointing to the deleted crossConnection instance shall be set to point to the Fabric responsible
for the connection of the termination points. The counters in the appropriate TP Pool objects (if
applicable) shall be updated. The connectivityPointer attributes in the disconnected termination
points shall be set to NULL. Deleting a crossConnection object instance has no effect on the
composition of any GTP*
;;
REGISTERED AS {etsNameBinding 16 };

etsCrossConnection-mpCrossConnection NAME BINDING
SUBORDINATE OBJECT CLASS "Recommendation M.3100":crossConnection
  AND SUBCLASSES;
NAMED BY
SUPERIOR OBJECT CLASS    "Recommendation M.3100":mpCrossConnection
  AND SUBCLASSES;
WITH ATTRIBUTE           "Recommendation M.3100":crossConnectionId;
BEHAVIOUR
  etsCrossConnection-mpCrossConnectionBehaviour BEHAVIOUR
     DEFINED AS
*The value of the fromTermination attribute in the  crossConnection object must be NULL. When an
instance of crossConnection is deleted, the following attributes will be affected. The
crossConnectionObjectPointer attributes in the termination points or in the gtp objects that were
pointing to the deleted crossConnection instance shall be set to point to the Fabric responsible
for the connection of the termination points. The counters in the appropriate TP Pool objects (if
applicable) shall be updated. The connectivityPointer attributes in the disconnected termination
points shall be set to NULL. Deleting the last cross-Connection contained in a multipoint cross
connection object has the effect of also deleting the multipoint cross connection object instance
(and updating the appropriate pointers). Deleting a crossConnection object instance has no effect
on the composition of any GTP*
;;
REGISTERED AS {etsNameBinding 17 };
```

9 Protection fragment

The protection fragment information model is to be found in ITU-T Recommendation G.774.03 [3].

9.1 Object classes

In this context the IMPORTS clause specifies the object classes which can be instantiated in the scope of this ETS. The IMPORTS clause does not include uninstantiated superclasses.

```
BEGIN
IMPORTS
protectedTTPBidirectional
protectedTTPSink
protectedTTPSource
protectionGroup
protectionUnit
sdhMSProtectionGroup
sdhMSProtectionUnit
unprotectedCTPBidirectional
unprotectedCTPSink
unprotectedCTPSource
FROM {ccitt(0) recommendation(0) g(7) g774(774) hyphen(127) prot(03) informationModel(0)
managedObjectClass(3) }
;
END
```

9.2 Packages

All packages associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

9.3 Attributes

All attributes associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

```
supportedByObjectList
    The value of the supportedByObjectList attribute points to the equipment and software
    objects which implement the TPs.
```

9.4 Name bindings

```
BEGIN
IMPORTS
protectedTTPBidirectional-sdhNE
protectedTTPSink-sdhNE
protectedTTPSource-sdhNE
augBidirectional-protectedTTPBidirectional
augSink-protectedTTPSink
augSource-protectedTTPSource
protectionGroup-managedElement
protectionUnit-protectionGroup
unprotectedCTPBidirectional-msTTPBidirectional
unprotectedCTPSink-msTTPSink
unprotectedCTPSource-msTTPSource
FROM {ccitt(0) recommendation(0) g(7) g774(774) hyphen(127) prot(03) informationModel(0)
nameBinding(6) }
;
END
```

10 Equipment fragment

10.1 Equipment - object classes

In this context, the IMPORTS clause specifies the object classes which can be instantiated in the scope of this ETS. The IMPORTS clause does not include uninstantiated superclasses.

```
BEGIN
IMPORTS
sdhNE
FROM {ccitt(0) recommendation(0) g(7) g774(774) informationModel(0) managedObjectClass(3) }

software
FROM M.3100ObjectClass {ccitt(0) recommendation(0) m(13) m3100(3100) informationModel(0)
managedObjectClass(3) }
;
END
```

The external TimePackage shall be supported by the sdhNE instance.

```
equipmentProtectionGroup MANAGED OBJECT CLASS
    DERIVED FROM     ”Recommendation G.774-03: 1994”:protectionGroup;
    CHARACTERIZED BY
        equipmentProtectionGroupPkg PACKAGE
            BEHAVIOUR equipmentProtectionGroupBeh;
            NOTIFICATIONS
                ”Recommendation G.774-03: 1994”: protectionSwitchReporting
equipmentProtectionStatusParameter;;;
REGISTERED AS {  etsObjectClass 14 };
equipmentProtectionGroupBeh BEHAVIOUR
    DEFINED AS
 ” This object class is used specifically for representing an equipment protection group in a
protection system.
The invokeProtection action switches always from protectedUnit to protectingUnit.
Either no or all equipmentProtectionUnits within an equipmentProtectionGroup shall have the
priorityPkg package.”;


equipmentProtectionUnit MANAGED OBJECT CLASS
    DERIVED FROM     ”Recommendation G.774-03: 1994”:protectionUnit;
    CHARACTERIZED BY
        ”Recommendation M.3100: 1992”:createDeleteNotificationsPackage,
        equipmentProtectionUnitPkg PACKAGE
            BEHAVIOUR equipmentProtectionUnitBeh;
            ATTRIBUTES
                equipmentProtectionStatus   GET;;;
REGISTERED AS { etsObjectClass 15 };
equipmentProtectionUnitBeh BEHAVIOUR
    DEFINED AS
 ” This object class is specific to equipment protection systems. Instances of this object class
are used to represent a relationship between a functional object and an equipment.
equipmentProtectionUnits are self-instanciated by the agent according to the equipment protection
schemes adopted by the NE. An equipmentProtectionUnit instance is deleted when the sdhEquipment
instance pointed to by the unreliableResourcePointer attribute are deleted, and is created again
if the sdhEquipment are instantiated with the appropriate equipmentExpected attribute value.
The reliableResourcePointer inherited by the protectionUnit superclass can be optionally an empty
set for instances of this object class.
The agent can also create and delete instances of the equipmentProtectionUnit in order to reflect
local modifications in the equipment protection schemes.
The attributeValueChange notification shall only be used to notify changes of the priority
attribute.”;


sdhEquipmentR MANAGED OBJECT CLASS
DERIVED FROM "Recommendation M.3100":equipment;
CHARACTERIZED BY
"Recommendation M.3100":administrativeOperationalStatesPackage,
"Recommendation M.3100":attributeValueChangeNotificationPackage,
"Recommendation M.3100":createDeleteNotificationsPackage,
"Recommendation M.3100":currentProblemListPackage,
"Recommendation M.3100":locationNamePackage,
"Recommendation M.3100":stateChangeNotificationPackage,
"Recommendation M.3100":equipmentEquipmentAlarmPackage,
"Recommendation X.721":availabilityStatusPackage,
sdhEquipmentPackage PACKAGE
BEHAVIOUR
sdhEquipmentBehaviour   BEHAVIOUR
 DEFINED AS
```

```
* The equipment object may be instantiated or exist without the presence of the physical
resources. In this case the operational state shall be "disabled" and the availability status
attribute shall contain the value "notInstalled".
When the resource is physically removed, the corresponding equipment object is not automatically
deleted.
The equipmentExpected attribute shall be provided at instantiation time. The create request shall
fail if the value of this attribute is unacceptable to the NE, and the failure reason shall
indicate this mismatch in the response.
When there is a mismatch in the contents of the equipmentActual and the equipment Expected
attribute, an equipmentAlarm notification with probable cause "replaceableUnitTypeMis
    match" shall be raised. This checking is only performed if the availabilityStatus does not
contain the value "notInstalled". The equipmentExpected value of "NULL" (no type) does not match
any other value than NULL for equipmentActual. Changes in the value of the equipmentExpected
attribute can only be achieved by object deletion and creation.
The "Recommendation M.3100":versionPackage package is not used.
The protectionUnitPointer attribute is used to point to a equipmentProtectionUnit within a
equipmentProtectionGroup if the equipment is protectet. Otherwise the Pointer is null. One
equipment could participate in several equipmentProtectionUnits at the same time.*
;;
ATTRIBUTES
"Recommendation M.3100":version GET,
equipmentExpected   GET,
equipmentActual GET,
specificPhysicalInstance    GET,
physicalConnectorList   GET,
protectionUnitPointer   GET;
;;
   REGISTERED AS { etsObjectClass 16 };
```

10.2 Equipment - attributes

All packages and attributes associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

```
affectedObjectList
The value of the affectedObjectList attribute represents the functional objects implemented by
the equipment or software object in which the attribute is applied. If the equipment or software
object become disabled, all managed objects referred to by the affectedObjectList shall also be
disabled.

equipmentActual             ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS12.EquipmentActual;
    BEHAVIOUR
    equipmentActualBehaviour BEHAVIOUR
    DEFINED AS
This attribute contains the equipment type of the equipment actually present. The "EquipmentType"
value is a vendor-specific identification of a particular set or class of equipment, where all
the set members have equivalent capability.
;;
REGISTERED AS {etsAttribute 4 };


equipmentExpected               ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS12.EquipmentExpected;
    BEHAVIOUR
        equipmentExpectedBehaviour BEHAVIOUR
    DEFINED AS
This attribute contains the equipment type requested at object creation. The "EquipmentType"
value is a vendor-specific identification of a particular set or class of equipment, where all
the set members have equivalent capability.
;;
REGISTERED AS {etsAttribute 5};


equipmentProtectionStatus ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS12.EquipmentProtectionStatus;
    MATCHES FOR         EQUALITY, SET-COMPARISON,
                        SET-INTERSECTION;
    BEHAVIOUR equipmentProtectionStatusBeh;
REGISTERED AS { etsAttribute 12 };
equipmentProtectionStatusBeh BEHAVIOUR
    DEFINED AS
 ″ This attribute is used to indicate the status of the protection switch in a protectionUnit
instance.
The remainder of this behaviour provides the allowable equipmentProtectionStatus attribute values
for protected and protecting units for both revertive and non-revertive systems.
The following allowable equipmentProtectionStatus values are associated with each protected unit:

No Request: No switch request is present on the unit.
Manual Switch to Protecting Complete: The unit has completed a Manual Switch.
Automatic Switch Pending: The unit has a failure condition present and the protecting unit is
unavailable.
```

Automatic Switch Complete: The unit has completed an Automatic Switch to the protecting unit due to failure condition.
Force Switch Complete, Automatic Switch Pending: The unit has completed a Force Switch. Additionally, the unit has an automatic switch pending.
Automatic Switch Complete, Wait-to-Restore: The unit has completed an Automatic Switch to the protecting unit. (revertive case only)
Force Switch Complete: The unit has completed a Force Switch to the protecting unit.
Protected Unit Lockout Completed: The unit has been locked out from the protecting unit.

Additionally the non-revertive protected protection unit has the following status values:

Do Not Revert: The protected unit has been switched to the protecting unit and the request to do so has been released. The switch to the protecting unit is maintained.

The following allowable protectionStatus values are associated with each protecting unit:

No Request: No request is present on the protecting unit.
Manual Switch to Protecting Unit Complete: The protected unit has completed a Manual Switch.
Automatic Switch Complete to Protecting Unit: The protected unit has completed an automatic switch to the protecting unit.
Automatic Switch Complete to Protecting Unit, Wait to Restore: The unit has completed an Automatic Switch to the protecting unit. (revertive case only)
Protecting Unit Failed: The protecting unit has a failure condition present.
Force Switch Complete to Protecting Unit: The unit has completed a Force Switch of a protected unit to the protecting unit.
Protecting Unit Locked Out: The protecting unit has been locked out.

Additionally, the non-revertive protecting unit has the following values:

Do Not Revert: The protected unit has been switched to the protecting unit and the request to do so has been released. The switch to the protecting unit is maintained.
Manual Switch to Protected Unit Complete: The unit has completed a Manual Switch from the protecting unit to the protected unit.
" ;

```
physicalConnectorList           ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS12.PhysicalConnectorList;
    BEHAVIOUR
        physicalConnectorListBehaviour BEHAVIOUR
    DEFINED AS
This attribute is used to relate external cabling to the appropriate transport objects. There is
an entry per connector.

;;
REGISTERED AS {etsAttribute 7 };


protectionUnitPointer       ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS12.ProtectionUnitPointer;
    MATCHES FOR             EQUALITY;
REGISTERED AS {etsAttribute 13 };


specificPhysicalInstance        ATTRIBUTE
    WITH ATTRIBUTE SYNTAX           PrETS12.PhysicalInstance;
    BEHAVIOUR
        specificPhysicalInstanceBehaviour BEHAVIOUR
    DEFINED AS
This attribute contains the unique identifier of the physical equipment (e.g. serial number).
This may be a manufacturer dependent serial numbes or other unique identifier (or unknownInstance
where the actual instance may not be determined from the actual equipment)

;;
REGISTERED AS {etsAttribute 6 };

[ for information only.
Version  ::=
-- Defined in M.3100, is used to present sufficient information to uniquely identify the
"equipmentActual" for the purpose of repair or reordering.
]
```

10.3 Equipment - parameter

```
equipmentProtectionStatusParameter PARAMETER
    CONTEXT EVENT-INFO;
    WITH SYNTAX             PrETS12.EquipmentProtectionStatusParameter;
    BEHAVIOUR equipmentProtectionStatusParameterBeh;
REGISTERED AS { etsParameter 1 };
equipmentProtectionStatusParameterBeh BEHAVIOUR
    DEFINED AS
”This parameter is included in the additional info parameters of the protection switching
reporting notification.
This notification is sent by the protection group according to the following rules. There are
several cases:
The switch from protected to protecting or protecting to protected has been done without
preempting an existing switch. In this case the old and new values of the Protection Status
attribute of the protecting channel shall be reported in the notification by means of the
oldProtectionStatus and newProtectionStatus parameters respectively.
A switch is perfomed by preemting an existing one. In this case the old and new values of the
Protection Status attribute of the protecting unit shall be reported in the notification by means
of the oldProtectionStatus and newProtectionStatus parameters respectively.
An autoswitch condition exists on an equipment but the auto-switch cannot be served due to the
unavailability of the equipment that otherwise protects it. In this case the oldProtectionStatus
and newProtectionStatus parameters refer to the protection status attribute value of the
equipment on which the autoswitch condition arises. The exception is when the equipment is
already forced or locked out, in which case no notification is sent.
A working equipment (protected unit) has been locked out or released from lockout without
modifying any existing switch. In this case the oldProtectionStatus and newProtectionStatus
parameters refer to the protection status attribute value of the working equipment which has been
locked out.
A protection equipment (protecting unit) has been lockout or released from lockout without
modifying the existing switch. In this case the oldProtectionStatus and newProtectionStatus
parameters refer to the protection status attribute value of the  protection equipment which has
been lockout.
The protectionSwitchReporting is not sent when the automatic switch condition is toggling between
the equipment failure and WTR condition. While in the lockout of forced switch state no
notification is sent except for ending of release failure.”;
```

10.4 Equipment - name bindings

All name bindings associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

```
BEGIN
IMPORTS
equipment-managedElement,
equipment-equipment
FROM {ccitt(0) recommendation(0) m(13) m3100(3100) informationModel(0) nameBinding(6) }
;
END
```

10.5 Equipment - supporting ASN.1

All ASN.1 types associated with object classes are implicitly imported from ITU-T Recommendations defining the appropriate object classes.

```
PrETS12{ccitt(0)identified-organization(4)etsi(0)ets304(304)informationModel(0)asn1Module(2)
prETS12(3)}
DEFINITIONS IMPLICIT TAGS ::=
BEGIN
-- EXPORTS everything

IMPORTS

Version FROM ASN1DefinedTypesModule {ccitt(0) recommendation(0) m(13) m3100(3100)
informationModel(0) asn1Module(2) asn1DefinedTypesModule(0)}
RDNSequence FROM InformationFramework {joint-iso-ccitt ds(5) modules(1) informationFramework(1)}
ObjectInstance FROM CMIP-1 {joint-iso-ccitt(2) ms(9) cmip(1) modules(0) protocol(3)}
SwitchStatus, FromAndToProtectionUnit FROM SDHProtASN1 {itu-t recommendation(0) g(7) g774(774)
hyphen(127) prot(03) informationModel(0) asn1Module(2) sdhmsp(0)};


Connector ::= SEQUENCE {
          connectorType     PrintableString,
          connectorLocation     PrintableString,
          supporting            ListOfLocalDistinguishedName}


EquipmentActual ::= CHOICE {
                noType              NULL,
                type                EquipmentType}
EquipmentAutoSwitchReason ::= CHOICE {
                            waitToRestore       [0] NULL,
                            equipmentFailure    [1] NULL
                                }


EquipmentExpected ::= CHOICE {
                noType              NULL,
                type                EquipmentType}


EquipmentProtectionStatus ::= SET OF CHOICE {
    noRequest                   [0] NULL,
    doNotRevert                 [1] NULL,
    manualSwitch                    [2] SEQUENCE {
        switchStatus                    [0] SwitchStatus,
        protectionUnitsSwitched         [1] FromAndToProtectionUnit},
    autoSwitch                  [3] SEQUENCE {
        switchStatus                    [0] SwitchStatus,
        protectionUnitsSwitched         [1] FromAndToProtectionUnit,
        autoSwitchReason                [2] EquipmentAutoSwitchReason},
    forcedSwitch                    [4] SEQUENCE {
        switchStatus                    [0] SwitchStatus,
        protectionUnitsSwitched         [1] FromAndToProtectionUnit},
    lockout                     [5] SwitchStatus}


EquipmentProtectionStatusParameter ::= SEQUENCE {
            oldProtectionStatus EquipmentProtectionStatus,
            newProtectionStatus EquipmentProtectionStatus}


EquipmentType ::= PrintableString


ListOfLocalDistinguishedName ::= SET OF RDNSequence


PhysicalConnectorList ::= SET OF Connector


PhysicalInstance ::= CHOICE {
                unknownInstance NULL,
                instance        PrintableString}


ProtectionUnitPointer ::= CHOICE  {
                pointer     SET OF ObjectInstance
                null        NULL }


END -- end of ASN1DefinedTypesModule
```

11 Support objects fragment

11.1 Support objects - object classes

```
powerSupply MANAGED OBJECT CLASS
DERIVED FROM    "Recommendation X.721":top;
CHARACTERIZED BY
"Recommendation M.3100":createDeleteNotificationsPackage,
powerSupplyPackage  PACKAGE
    BEHAVIOUR
   powerSupplyBehaviourPkg BEHAVIOUR
    DEFINED AS
The power supply object class is used to control the power supply sources within the SDHNE. There
shall be one instance for each of the power supply sources.
;
;
ATTRIBUTES
    powerSupplyId       GET,
    powerSource     GET,
    poweredEquipmentPtrList     GET,
    "Recommendation X.721":operationalState     GET,
    "Recommendation M.3100":supportedByObjectList       GET;
NOTIFICATIONS
    "Recommendation X.721":attributeValueChange,
    "Recommendation X.721":stateChange;
;;
REGISTERED AS { etsObjectClass 11 };
syncProtectionGroup MANAGED OBJECT CLASS
        DERIVED FROM    "Recommendation G.774-03: 1994":protectionGroup;
        CHARACTERIZED BY
                syncProtectionGroupPkg PACKAGE
                        BEHAVIOUR syncProtectionGroupBeh;
                        NOTIFICATIONS
                                "Recommendation G.774-03: 1994":protectionSwitchReporting
                                syncProtectionStatusParameter;;;
        CONDITIONAL PACKAGES
    squelchStatusPkg    PRESENT IF
        *an instance represents the T4 selection function*,
    squelchThresholdsPkg    PRESENT IF
        *an instance represents the T4 selection function and the SSM algorithm is
        supported*,
    sSMActivePackage    PRESENT IF
        *activation/deactivation of the SSM algorithm is supported*;
REGISTERED AS { etsObjectClass 17 };
syncProtectionGroupBeh BEHAVIOUR
        DEFINED AS
"This object class is specific to synchronisation protection. The creation/deletion of a
syncProtectionGroup is performed by the NE depending upon its synchronisation protection
capabilities (intrinsic knowledge of the NE). The operationalState attribute reflects only the
switch capabilities of the protection scheme(i.e. it becomes disabled when the NE has detected
that it can no longer perform switch operations (automatic or manual)).
The manual switch operation can be performed among timing sources with the same or different
priority and, when the SSM algorithm is active, with the same quality. A manual switch request
between two timing sources with different quality levels and SSM algorithm is active will be
rejected.
The forced switch operation can be performed among any timing sources, independently of the
priority and quality levels.
All protectionUnits work in the protecting mode.
The protectionSwitchReporting notification is sent by the synchronisation protection group
according to the following rules. There are several cases:
A switch from one protecting unit to another has occurred. In this case the new values of the
protection status attributes of the protecting units which have been switched to and from shall
be reported in the notification by means of the toProtectionStatus and fromProtectionStatus
parameters respectively.
A protecting unit has a failure or a clearing of a failure without causing a switch. In this case
the fromProtectionStatus parameter is not used and the toProtectionStatus parameter refers to the
protection status attribute value of the protecting unit which has a failure or a clearing of a
failure. The exception is when the protecting unit is already forced or locked out, in which case
no notification is sent.
A protecting unit has been locked out or released from lockout without causing a switch. In this
case the fromProtectionStatus parameter is not used and the toProtectionStatus parameter refers
to the protection status attribute value of the protecting unit which has been locked out or
released from lockout.
A protecting unit has been manually switch or released from manual switch without causing a
switch. In this case the fromProtectionStatus parameter is not used and the toProtectionStatus
parameter refers to the protection status attribute value of the protecting unit which has been
manually switch or released from manual switch.
A protecting unit has been forced switch or released from forced switch without causing a switch.
In this case the fromProtectionStatus parameter is not used and the toProtectionStatus parameter
refers to the protection status attribute value of the protecting unit which has been forced
switch or released from forced switch.
While in the lockout or forced switch state, no notifications are sent ”;
```

```
syncProtectionUnit MANAGED OBJECT CLASS
        DERIVED FROM    "Recommendation G.774-03: 1994":protectionUnit;
        CHARACTERIZED BY
                "Recommendation M.3100: 1992":createDeleteNotificationsPackage,
                syncProtectionUnitPkg PACKAGE
                        BEHAVIOUR syncProtectionUnitBeh;
                        ATTRIBUTES
                                syncProtectionStatus    GET;;;
        CONDITIONAL PACKAGES
                sSMQualityPackage                       PRESENT IF
                        *the timing source has a SSM quality level*;
REGISTERED AS { etsObjectClass 18 };
syncProtectionUnitBeh BEHAVIOUR
        DEFINED AS
"This object class is specific to synchronisation protection. Instances of this object class are
used to represent a relationship between a timing source and the timing generator(or a timing
physical termination). All instances of this object class will act in the protecting role (i.e.
there will not be a syncProtectionUnit with the protecting attribute set to FALSE), in order to
ensure the consistency of the priority list.
This object class can be created by the OS or the NE. When it is created by the OS, the OS must
provide a value for the attribute unreliableResourcePointer.";

timingGenerator MANAGED OBJECT CLASS
   DERIVED FROM "Recommendation X.721":top;
   CHARACTERIZED BY
timingGeneratorPackage  PACKAGE
    BEHAVIOUR
   timingGeneratorBehaviourPkg BEHAVIOUR
       DEFINED AS
For the selection of the timing sources a 1:n protection like mechanism is used.
Each protectionUnit in the protectionGroup has a pointer, the unreliableResourcePointer, which
points to the related TP (see figure A.15).

The currentTimingSourcePointer points to the timing source currently in use. A value of NULL of
this attribute indicates the use of the internal oscillator. In that case, the
unreliableResourcePointer of the corresponding protectionUnit also points to NULL.
The reliableResourcePointer of the protectionUnit related to the currently used timing source
points to the timingGenerator. The reliableResourcePointers of the other protectionUnits related
to timing sources which are not currently in use are pointing to NULL.
To select a special instance of a possible timing source, the OS has to use the invokeProtection
action of the protectionGroup.
Only one instance of this object class shall be created.
;;
ATTRIBUTES
    timingGeneratorId       GET,
    currentTimingSourcePointer      GET,
    "Recommendation X.721":operationalState     GET,
    "Recommendation M.3100":supportedByObjectList       GET;
NOTIFICATIONS
    "Recommendation X.721":attributeValueChange,
    "Recommendation X.721":stateChange;
;;
REGISTERED AS { etsObjectClass 12 };


timingPhysicalTerminationBidirectional      MANAGED OBJECT CLASS
        DERIVED FROM    timingPhysicalTerminationSink,
                                timingPhysicalTerminationSource;
        CHARACTERIZED BY
                timingPhysicalTerminationBidirectionalPkg PACKAGE
                        BEHAVIOUR timingPhysicalTerminationBidirectionalBehaviour;;;
REGISTERED AS { etsObjectClass 19 };
timingPhysicalTerminationBidirectionalBehaviour       BEHAVIOUR
        DEFINED AS
"This object class originates and terminates the 2 MHz signal.";

timingPhysicalTerminationSink     MANAGED OBJECT CLASS
        DERIVED FROM    "Recommendation X.721: 1992":top;
        CHARACTERIZED BY
                timingPhysicalTerminationSinkPkg        PACKAGE
                        BEHAVIOUR timingPhysicalTerminationSinkBehaviour;
                        ATTRIBUTES
                                timingPhysicalTerminationId                             GET,
                                "Recommendation X.721: 1992":operationalState           GET,
                                "Recommendation M.3100: 1992":supportedByObjectList   GET;
                        NOTIFICATIONS
                                "Recommendation X.721: 1992":objectCreation,
                                "Recommendation X.721: 1992":objectDeletion,
                                "Recommendation X.721: 1992":stateChange,
                                "Recommendation X.721: 1992":attributeValueChange;;;
        CONDITIONAL PACKAGES
                "Recommendation M.3100: 1992":tmnCommunicationsAlarmInformationPackage
                        PRESENT IF  "There are alarms associated with the
timingPhysicalTermination",
```

```
                "Recommendation M.3100: 1992":alarmSeverityAssignmentPointerPackage
                        PRESENT IF  "an instance supports it";
REGISTERED AS { etsObjectClass 20 };
timingPhysicalTerminationSinkBehaviour      BEHAVIOUR
        DEFINED AS
"This managed object represents the external 2 MHz input interface (reference point T3 in the
Recommendation G.783). This object class terminates (receives) the 2 MHz signal. The probable
causes which are reported by a communicationsAlarm notification are lossOfSignal and
degradedSignal.";

timingPhysicalTerminationSource MANAGED OBJECT CLASS
    DERIVED FROM        "Recommendation X.721: 1992": top;
    CHARACTERIZED BY
    timingPhysicalTerminationSourcePkg  PACKAGE
        BEHAVIOUR   timingPhysicalTerminationSourceBeh;
        ATTRIBUTES
            timingPhysicalTerminationId                 GET,
            "Recommendation X.721: 1992":operationalState       GET,
            "Recommendation M.3100: 1992":supportedByObjectList GET,
            outputTimingSourcePointer                   GET;
        NOTIFICATIONS
            "Recommendation X.721: 1992":objectCreation,
            "Recommendation X.721: 1992":objectDeletion,
            "Recommendation X.721: 1992":stateChange,
            "Recommendation X.721: 1992":attributeValueChange;;;
REGISTERED AS {etsObjectClass 21};
timingPhysicalTerminationSourceBeh      BEHAVIOUR
DEFINED AS
"This managed object  represents the external 2 MHz output interface (reference point T4 in the
Recommendation G.783). This object class originates (transmits) the 2 MHz signal.";
```

11.2 Support objects - packages

```
sSMActivePackage PACKAGE
        ATTRIBUTES
                sSMActive   GET-REPLACE;
REGISTERED AS { etsPackage 1 };

sSMQualityPackage PACKAGE
        ATTRIBUTES
                sSMQuality      GET-REPLACE;
REGISTERED AS { etsPackage 2 };

squelchStatusPkg PACKAGE
    ATTRIBUTES
        squelchStatus       GET;
REGISTERED AS { etsPackage 3 };
squelchThresholdsPkg PACKAGE
    ATTRIBUTES
        squelchT0Threshold      GET-REPLACE,
        squelchT1Threshold      GET-REPLACE;
REGISTERED AS { etsPackage 4 };
```

11.3 Support objects - attributes

```
supportedByObjectList
The value of the supportedByObjectList attribute points to the equipment and software objects
which implement the TPs.

currentTimingSourcePointer   ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.CurrentTimingSourcePointer;
    MATCHES FOR EQUALITY;
REGISTERED AS {etsAttribute 10 };

outputTimingSourcePointer       ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.OutputTimingSourcePointer;
    MATCHES FOR             EQUALITY;
REGISTERED AS {etsAttribute 14 };

poweredEquipmentPtrList   ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.PoweredEquipmentPtrList;
    MATCHES FOR EQUALITY, SET-COMPARISON, SET-INTERSECTION;
   BEHAVIOUR
      poweredEquipmentPtrListBehaviour BEHAVIOUR
      DEFINED AS
         This attribute is used to point to the equipment object instances which are powered by a
powerSupply instance.
;;
REGISTERED AS {etsAttribute 11 };

powerSource ATTRIBUTE
   WITH ATTRIBUTE SYNTAX       PrETS13.PowerSource ;
   MATCHES FOR EQUALITY;
   BEHAVIOUR
      powerSourceBehaviour BEHAVIOUR
      DEFINED AS
         This attribute is used to display the voltage of a power source.
;;
REGISTERED AS { etsAttribute 9 };

powerSupplyId ATTRIBUTE
   WITH ATTRIBUTE SYNTAX       PrETS13.NameType ;
   MATCHES FOR                 EQUALITY;
   BEHAVIOUR
      powerSupplyIdBehaviour BEHAVIOUR
      DEFINED AS
         This attribute is used as an RDN for naming instances of the powerSupply object classes.
;;
REGISTERED AS { etsAttribute 8 };

sSMActive ATTRIBUTE
  WITH ATTRIBUTE SYNTAX       PrETS13.Boolean;
  MATCHES FOR     EQUALITY;
  BEHAVIOUR
    sSMActiveBehaviour BEHAVIOUR
    DEFINED AS
"This attribute indicates whether or not the SSM algorithm is active(active = TRUE).";;
REGISTERED AS { etsAttribute 15 };

sSMQuality ATTRIBUTE
  WITH ATTRIBUTE SYNTAX       PrETS13.Integer;
  MATCHES FOR     EQUALITY;
  BEHAVIOUR
    sSMQualityBehaviour BEHAVIOUR
    DEFINED AS
"This attribute indicates the quality(0 to 15) of the timing source. Set operations on this
attribute are not allowed for interfaces that  carry quality information. When active, the SSM
algorithm gives priority to the quality indications before considering priority indications
attached to the protection units.";;
REGISTERED AS { etsAttribute 16 };

squelchStatus ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.Boolean;
    MATCHES FOR EQUALITY;
    BEHAVIOUR
        squelchStatusBehaviour BEHAVIOUR
        DEFINED AS
    ”This attribute indicates whether the selected input (either T0 or T1) to the T4 selection
proces is squelched or not (squelched = TRUE).”;;
REGISTERED AS { etsAttribute 17 };

squelchT0Threshold ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.Integer;
    MATCHES FOR EQUALITY;
    BEHAVIOUR
        squelchT0ThresholdBehaviour BEHAVIOUR
        DEFINED AS
```

```
    ”This attribute indicates the quality (0..15) of the T0 output below which the T0 output is
squelched ”;;
REGISTERED AS { etsAttribute 18 };

squelchT1Threshold ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.Integer;
    MATCHES FOR EQUALITY;
    BEHAVIOUR
        squelchT1ThresholdBehaviour BEHAVIOUR
        DEFINED AS
    ”This attribute indicates the quality (0..15) of the T1 reference selected by selector A
below which the T1 reference is squelched ”;;
REGISTERED AS { etsAttribute 19 };
syncProtectionStatus ATTRIBUTE
        WITH ATTRIBUTE SYNTAX       PrETS13.SyncProtectionStatus;
        MATCHES FOR             EQUALITY, SET-COMPARISON,
                                          SET-INTERSECTION;
        BEHAVIOUR syncProtectionStatusBeh;
REGISTERED AS { etsAttribute 20 };
syncProtectionStatusBeh BEHAVIOUR
       DEFINED AS
"This attribute is used to indicate the status of the protection switch in a protectionUnit
instance.
The remainder of this behaviour provides the allowable syncProtectionStatus attribute values for
protecting units.
No Request: No request is present on the protecting unit.
Manual Switch Complete to Protecting Unit: A Manual Switch has been completed to this protecting
unit.
Force Switch Complete to Protecting Unit: A Force Switch has been completed to this protecting
unit.
Automatic Switch Complete to Protecting Unit: An Automatic Switch has been completed to this
protecting unit.
Protecting Unit Failed: The protecting unit has a failure condition present.
Protecting Unit Locked Out: The protecting unit has been locked out.";

timingGeneratorId ATTRIBUTE
   WITH ATTRIBUTE SYNTAX       PrETS13.NameType ;
   MATCHES FOR                 EQUALITY;
   BEHAVIOUR
      timingGeneratorIdBehaviour BEHAVIOUR
      DEFINED AS
         This attribute is used as an RDN for naming instances of the timingGenerator object
classes.
;;
REGISTERED AS { etsAttribute 22 };

timingPhysicalTerminationId     ATTRIBUTE
    WITH ATTRIBUTE SYNTAX   PrETS13.NameType;
    MATCHES FOR             EQUALITY;
        BEHAVIOUR
                timingPhysicalTerminationIdBehaviour    BEHAVIOUR
                DEFINED AS
"The timingPhysicalTerminationId is an attribute type whose distinguished value can be used as an
RDN when naming an instance of the timingPhysicalTermination object class.";;
REGISTERED AS {etsAttribute 21 };
```

11.4 Support objects - name bindings

```
powerSupply-sdhNE NAME BINDING
   SUBORDINATE OBJECT CLASS powerSupply;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774" : sdhNE;
   WITH ATTRIBUTE           powerSupplyId;
   BEHAVIOUR
        powerSupply-sdhNEBehaviour   BEHAVIOUR
    DEFINED AS
    The subordinate managed object is automatically instantiated when the superior managed object
is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 18 };

syncProtectionUnit-syncProtectionGroup NAME BINDING
  SUBORDINATE OBJECT CLASS syncProtectionUnit;
  NAMED BY SUPERIOR OBJECT CLASS syncProtectionGroup;
  WITH ATTRIBUTE "Recommendation G.774-03:1994":protectionUnitId;
  CREATE
    WITH-AUTOMATIC-INSTANCE-NAMING;
  DELETE
    ONLY-IF-NO-CONTAINED-OBJECTS;
  REGISTERED AS { etsNameBinding 26 };

timingGenerator-sdhNE NAME BINDING
   SUBORDINATE OBJECT CLASS timingGenerator;
   NAMED BY
   SUPERIOR OBJECT CLASS    "Recommendation G.774":sdhNE;
   WITH ATTRIBUTE           timingGeneratorId;
   BEHAVIOUR
    timingGenerator-sdhNEBehaviour   BEHAVIOUR
    DEFINED AS
    The subordinate managed object is automatically instantiated when the superior managed object
is instantiated, according to the make-up and mode of operation of the equipment.
;;
REGISTERED AS { etsNameBinding 19 };

timingPhysicalTerminationBidirectional-sdhNE    NAME BINDING
        SUBORDINATE OBJECT CLASS        timingPhysicalTerminationBidirectional;
        NAMED BY
        SUPERIOR OBJECT CLASS       "Recommendation G.774":sdhNE;
        WITH ATTRIBUTE          timingPhysicalTerminationId;
        BEHAVIOUR timingPhysicalTerminationBidirectional-sdhNEBeh BEHAVIOUR
DEFINED AS
"The subordinate managed object is automatically instantiated when the superior managed object is
instantiated, according to the make-up and mode of operation of the equipment.";;
REGISTERED AS { etsNameBinding 27 };

timingPhysicalTerminationSink-sdhNE   NAME BINDING
        SUBORDINATE OBJECT CLASS        timingPhysicalTerminationSink;
        NAMED BY
        SUPERIOR OBJECT CLASS       "Recommendation G.774":sdhNE;
        WITH ATTRIBUTE          timingPhysicalTerminationId;
        BEHAVIOUR timingPhysicalTerminationSink-sdhNEBeh BEHAVIOUR
DEFINED AS
"The subordinate managed object is automatically instantiated when the superior managed object is
instantiated, according to the make-up and mode of operation of the equipment.";;
REGISTERED AS { etsNameBinding 28 };

timingPhysicalTerminationSource-sdhNE NAME BINDING
    SUBORDINATE OBJECT CLASS        timingPhysicalTerminationSource;
    NAMED BY SUPERIOR OBJECT CLASS      "Recommendation G.774":sdhNE;
    WITH ATTRIBUTE              timingPhysicalTerminationId;
    BEHAVIOUR   timingPhysicalTerminationSource-sdhNEBeh BEHAVIOUR DEFINED AS
                "The subordinate managed object is automatically instantiated when the superior
managed object is instantiated, according to the make-up and mode of operation of the
equipment.";;
REGISTERED AS {etsNameBinding 29 };
```

11.5 Support objects - parameter

```
syncProtectionStatusParameter     PARAMETER
        CONTEXT EVENT-INFO;
        WITH SYNTAX         PrETS13.SyncProtectionStatusParameter;
        BEHAVIOUR syncProtectionStatusParameterBeh;
REGISTERED AS { etsParameter 2 };
syncProtectionStatusParameterBeh BEHAVIOUR
        DEFINED AS
"This parameter is included in the additional info parameters of the protection switching
reporting notification.";
```

11.6 Support objects - supporting ASN.1

```
PrETS13 {ccitt(0) identified-organization(4) etsi(0) ets304(304) informationModel(0)
asn1Module(2) prETS13(4)}

DEFINITIONS IMPLICIT TAGS ::=
BEGIN
-- EXPORTS everything

IMPORTS

NameType FROM ASN1DefinedTypesModule {ccitt(0) recommendation(0) m(13) m3100(3100)
informationModel(0) asn1Module(2) asn1DefinedTypesModule(0)}

ObjectInstance FROM CMIP-1 {joint-iso-ccitt ms(9) cmip(1) modules(0) protocol(3)
RelativeDistinguishedName};


Boolean ::= BOOLEAN

CurrentTimingSourcePointer ::= CHOICE {
                pointer         [0] ObjectInstance,
                internalOscillator  [1] NULL
        }
Integer ::= INTEGER

OutputTimingSourcePointer ::=   ObjectInstance

PoweredEquipmentPtrList ::= SET OF ObjectInstance

PowerSource ::= INTEGER

SyncProtectionStatus ::= CHOICE {
    noRequest   [0] NULL,
    autoSwitch  [1] RelativeDistinguishedName,
    manualSwitch    [2] RelativeDistinguishedName,
    forcedSwitch    [3] RelativeDistinguishedName,
    lockout     [4] NULL,
    failure     [5] NULL
}
-- In the SyncProtectionStatus syntax, autoSwitch, manualSwitch and forcedSwitch will contain the
RDN of the synchronisation protection unit which has been switched from.

SyncProtectionStatusParameter ::= SEQUENCE {
    toProtectionStatus  SyncProtectionStatus,
    fromProtectionStatus    SyncProtectionStatus OPTIONAL}

END
```

Annex A (normative): Figures and tables

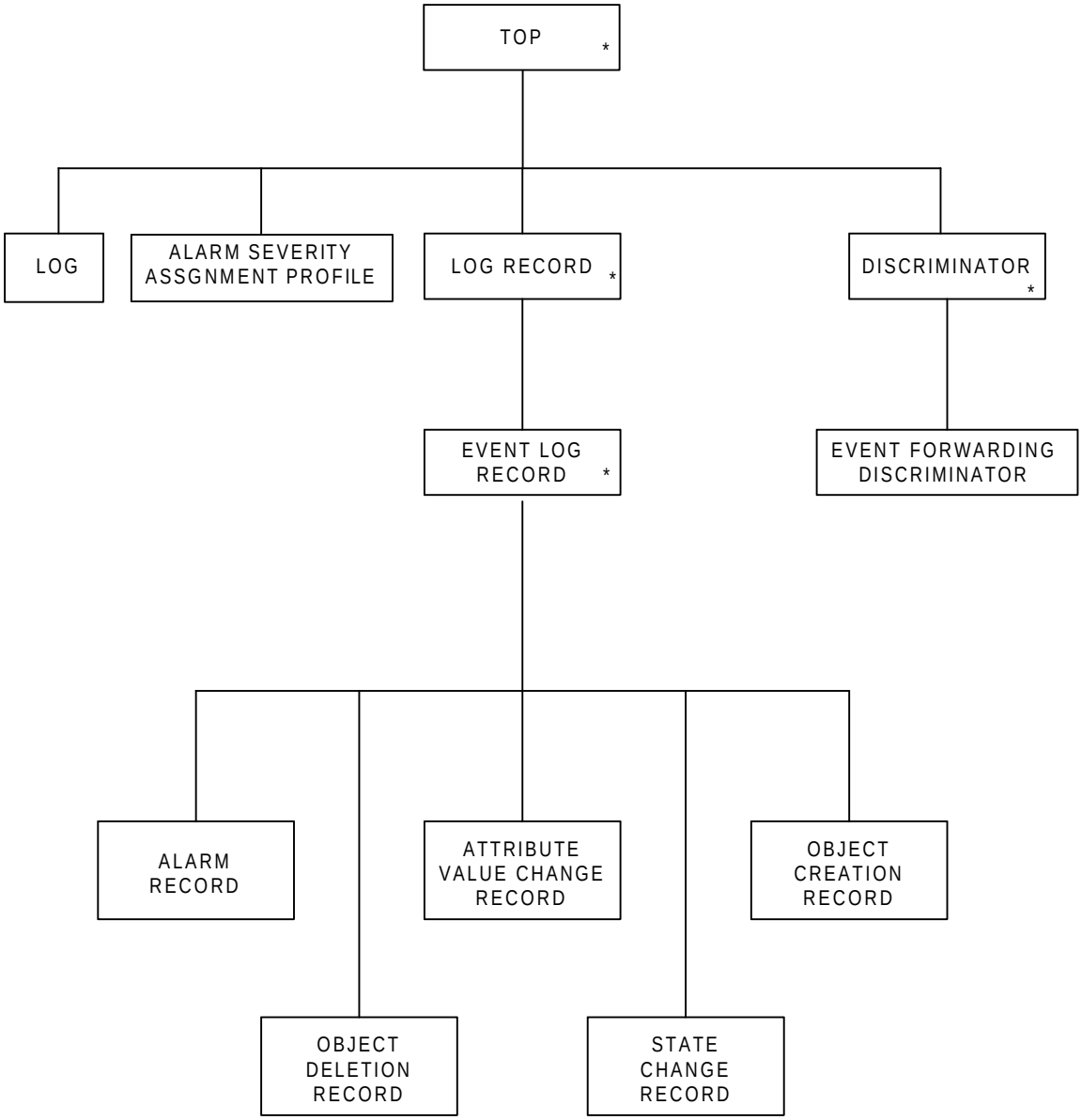


Figure A.1: Generic objects inheritance

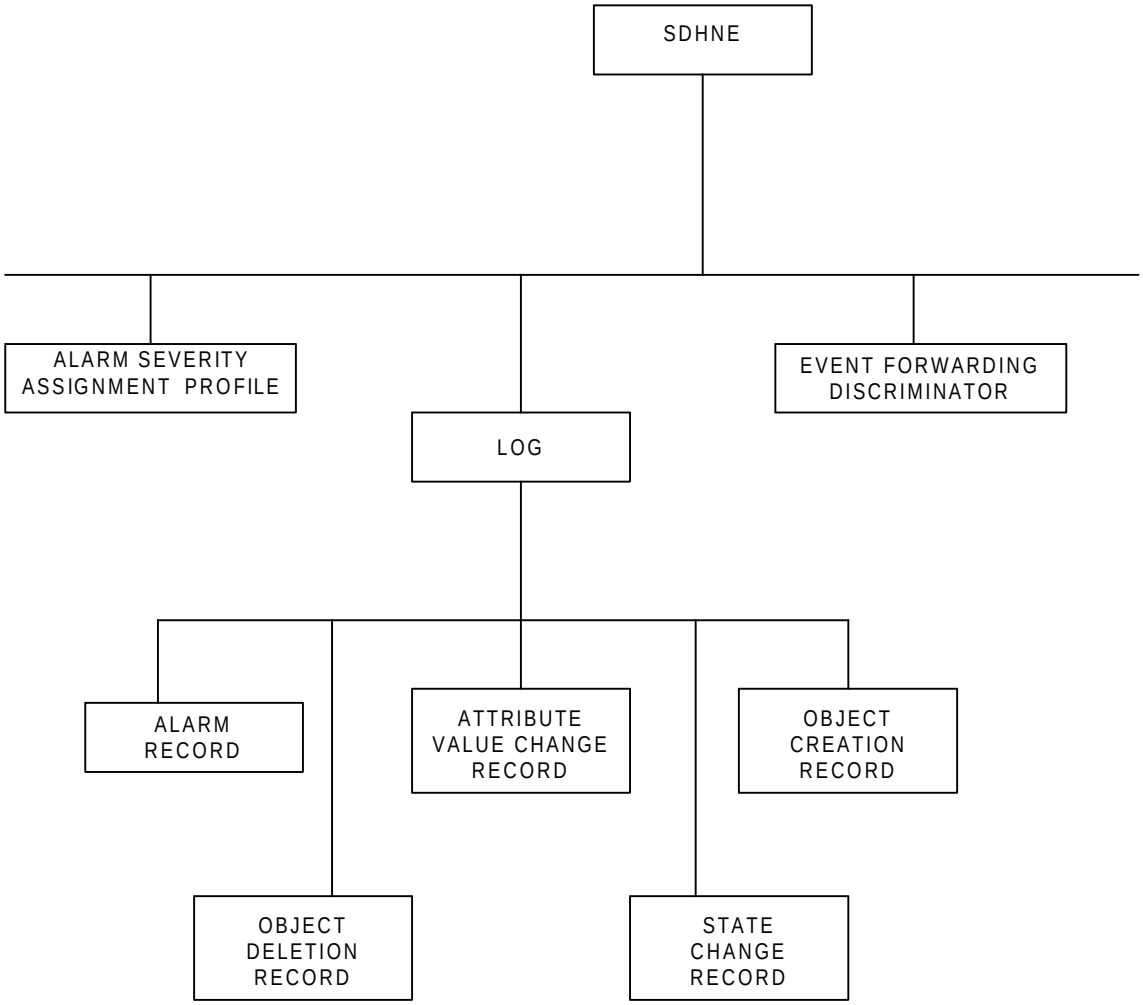


Figure A.2: Generic object naming

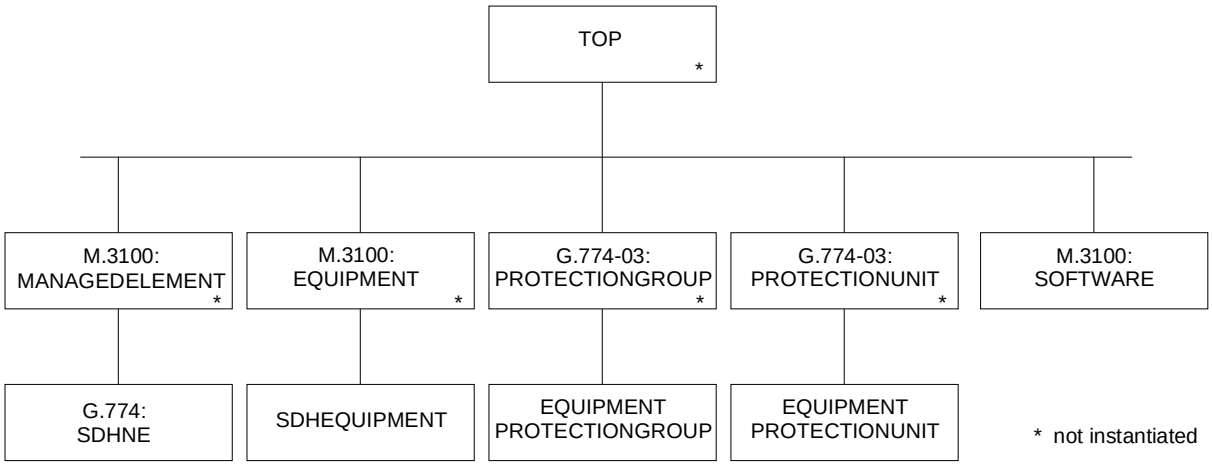


Figure A.3: Equipment objects inheritance

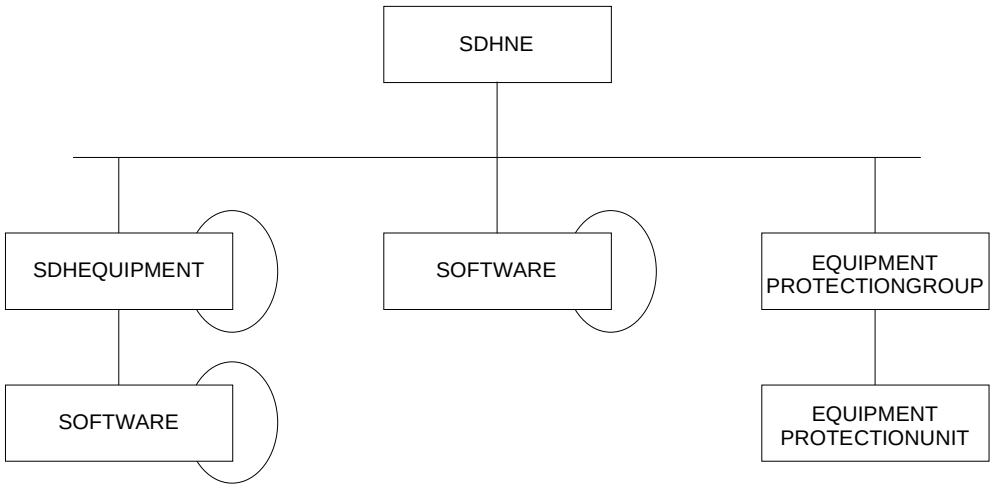


Figure A.4: Equipment object naming

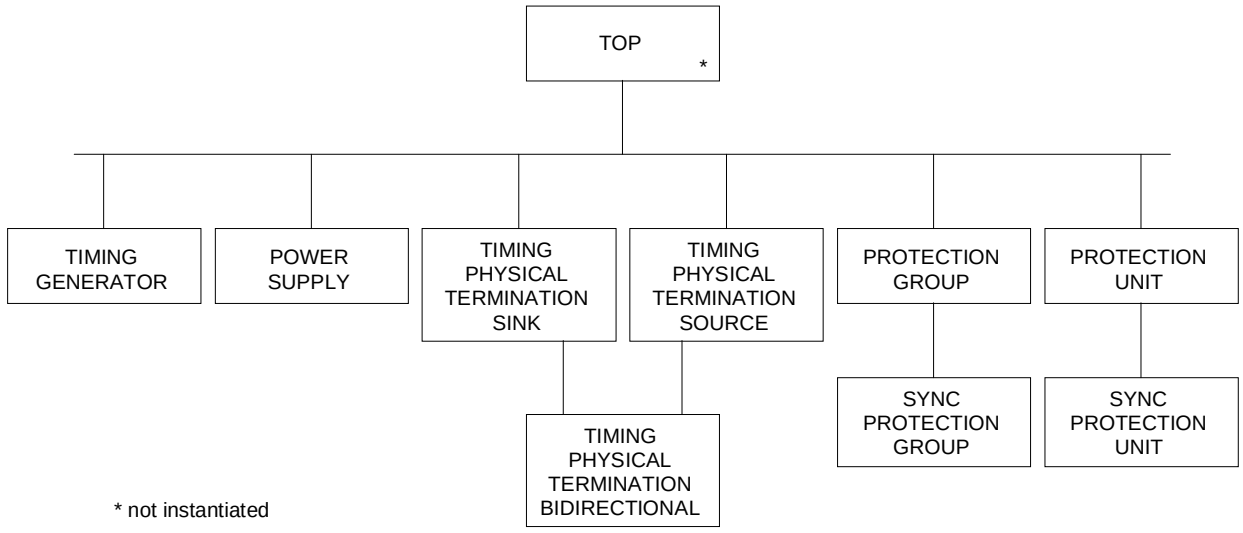


Figure A.5: Support objects inheritance

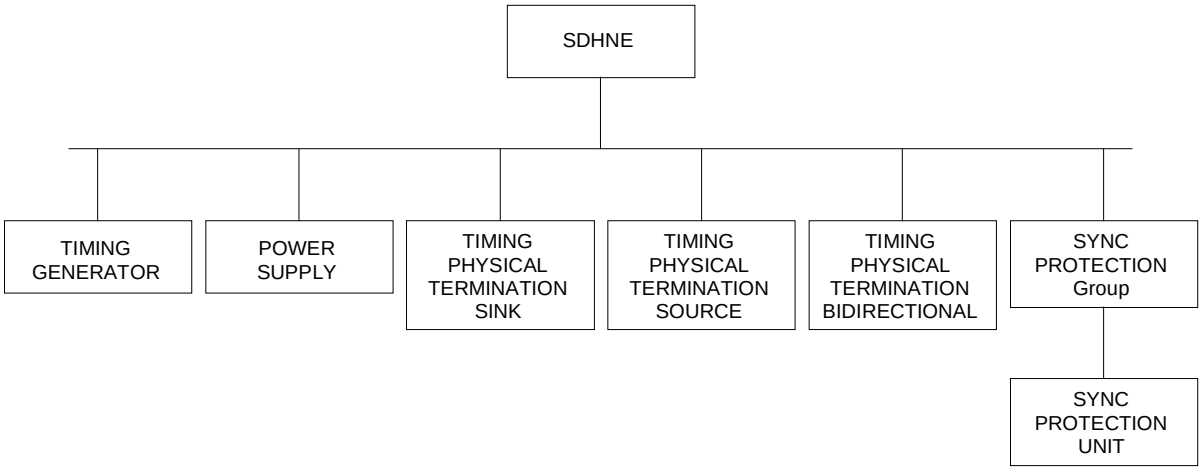
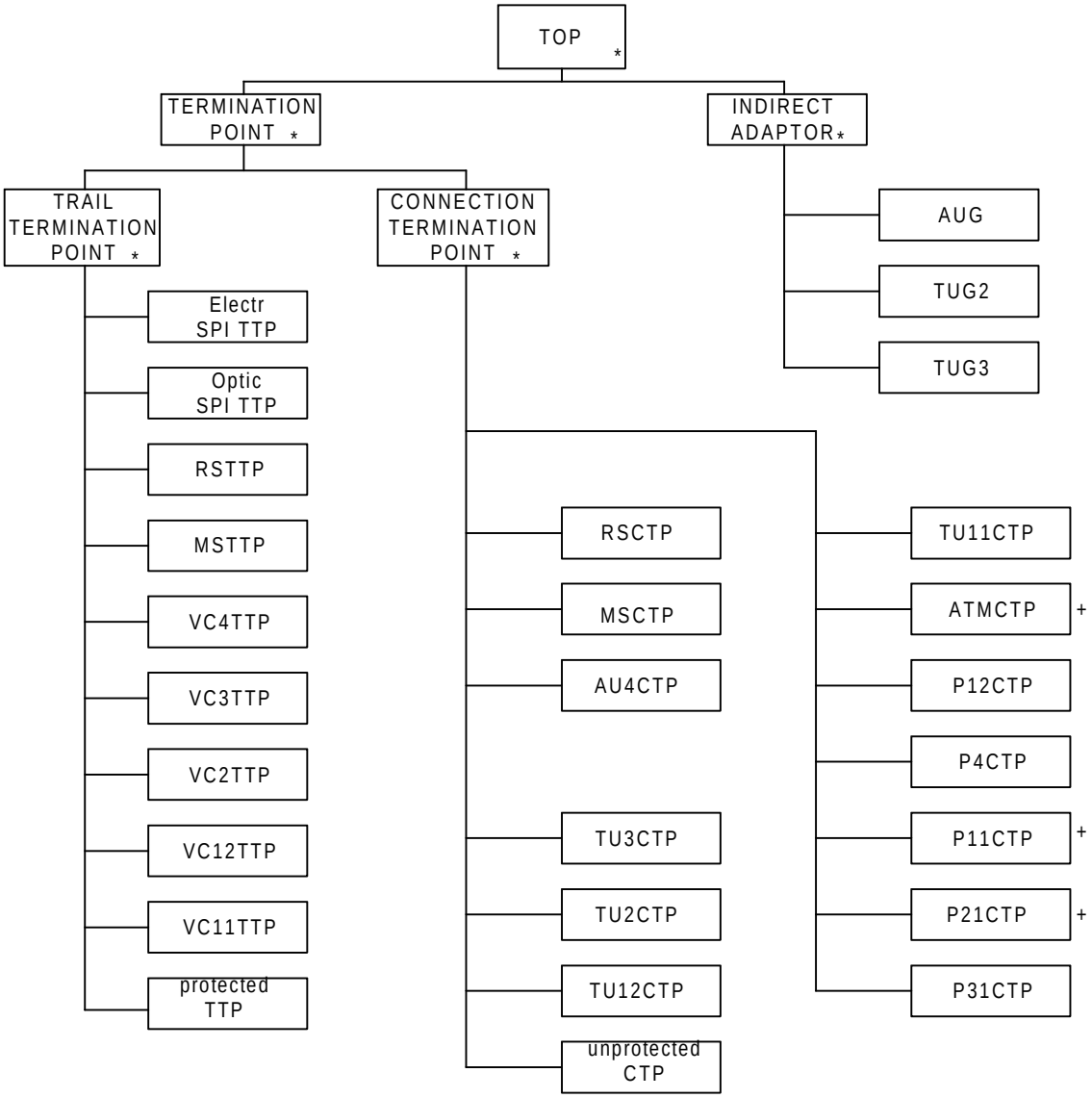


Figure A.6: Support objects naming

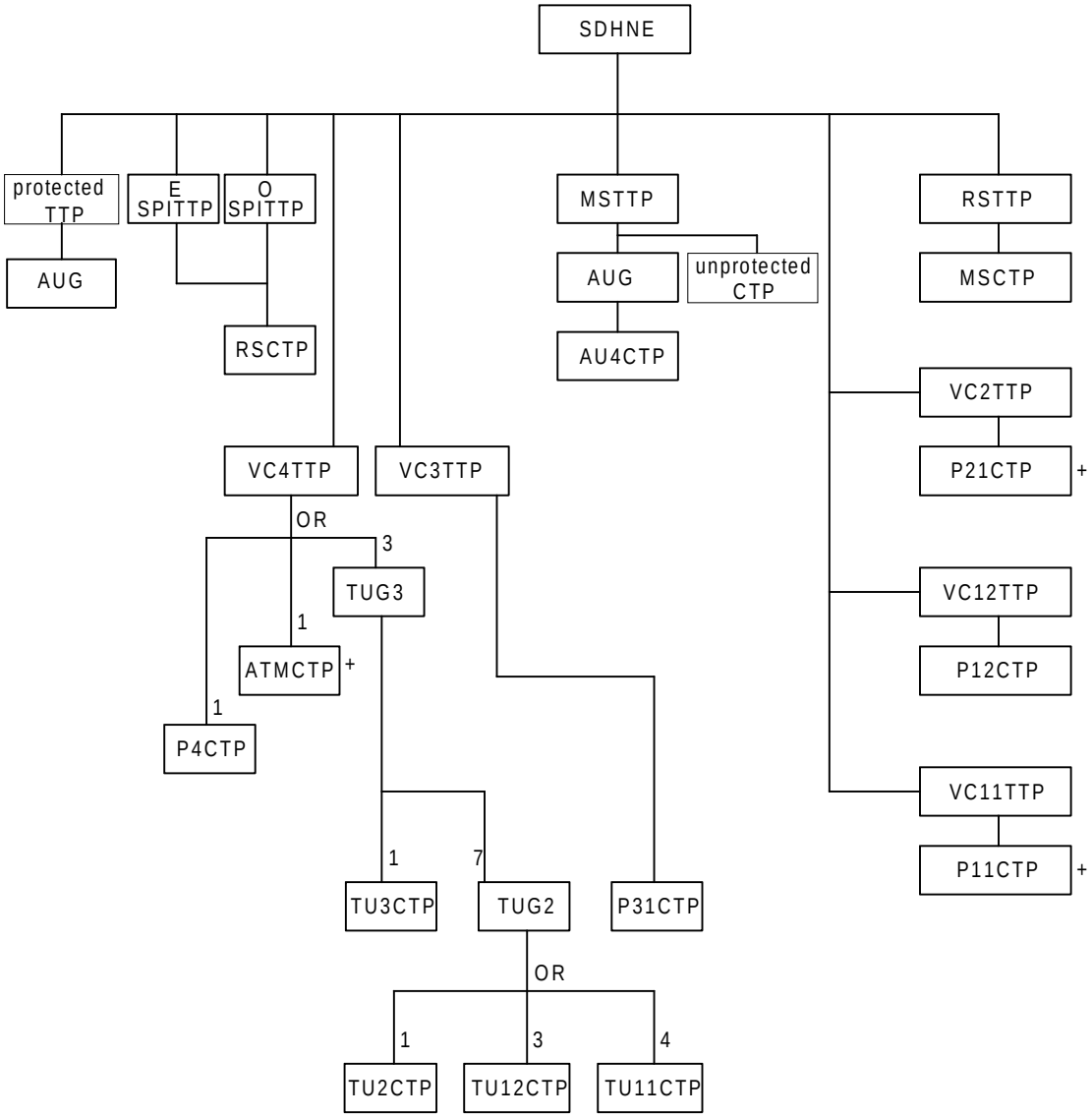


(All objects may be source, sink or bidirectional)

* not instantiated

\+ not defined

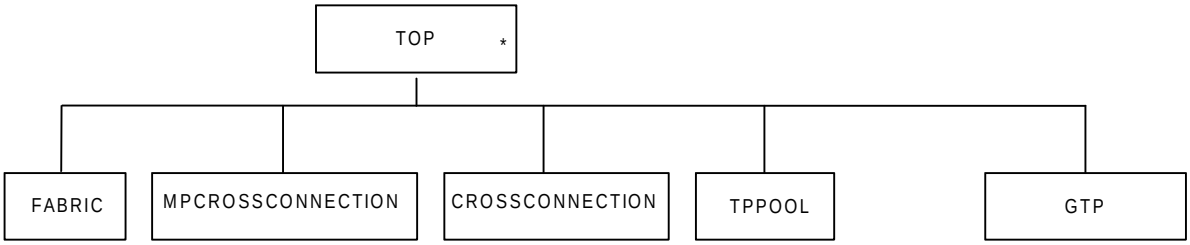
Figure A.7: Transport objects inheritance



(All objects may be source, sink or bidirectional)

\+ not defined

Figure A.8: Transport objects naming



* not instantiated

Figure A.9: Cross-connection objects inheritance

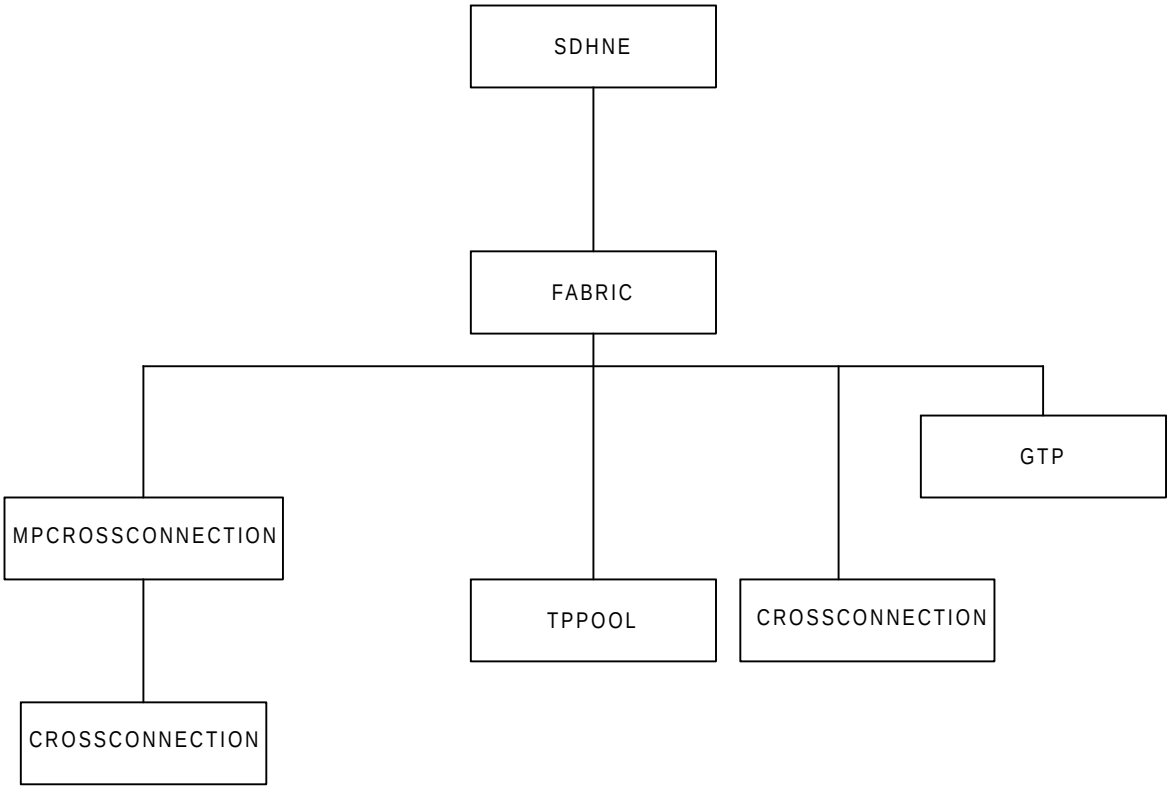
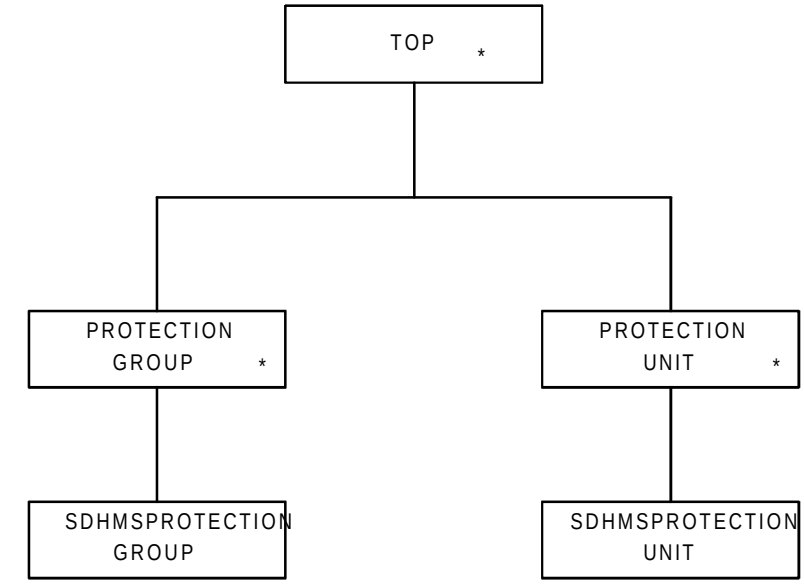


Figure A.10: Cross-connection objects naming



* not instantiated

Figure A.11: Section protection objects inheritance

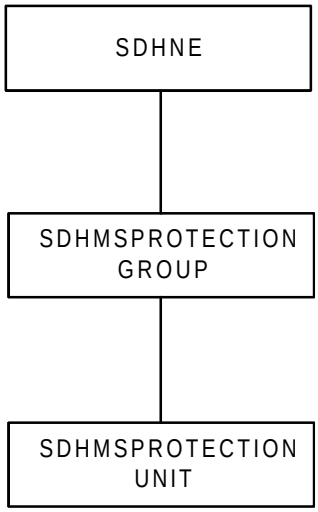


Figure A.12: Section protection objects naming

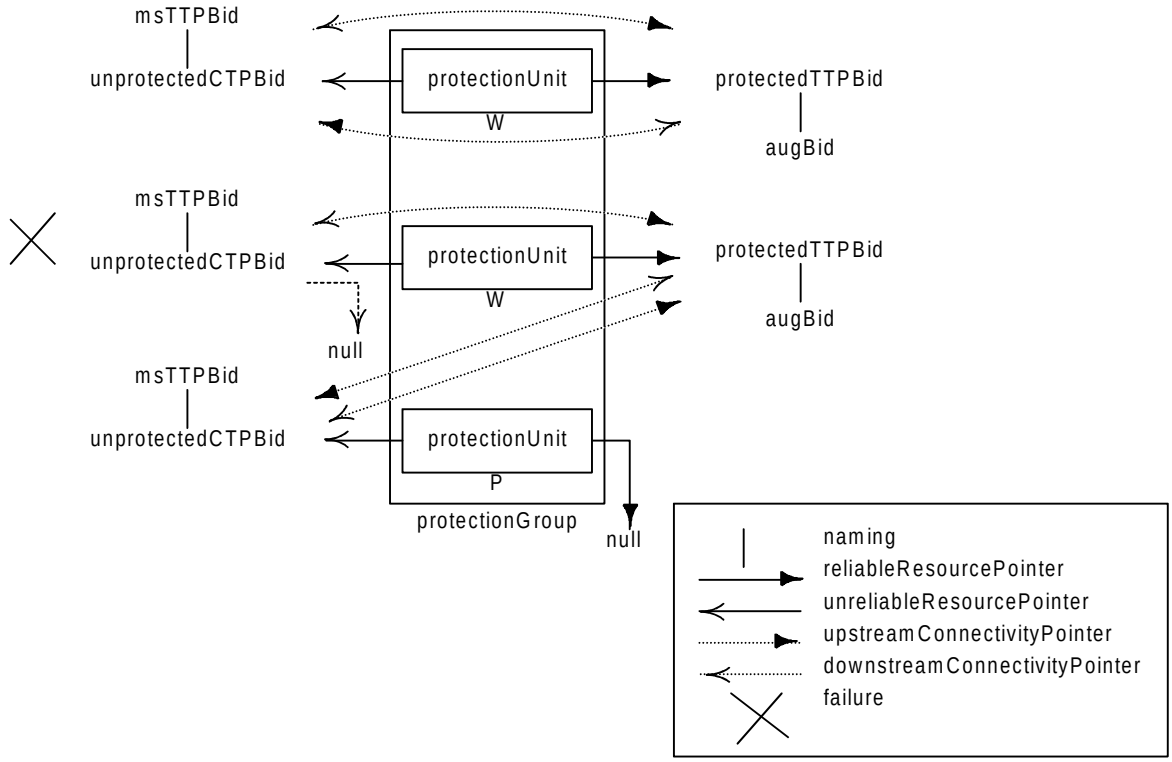


Figure A.13: Bi-directional 1:2 Protection model example

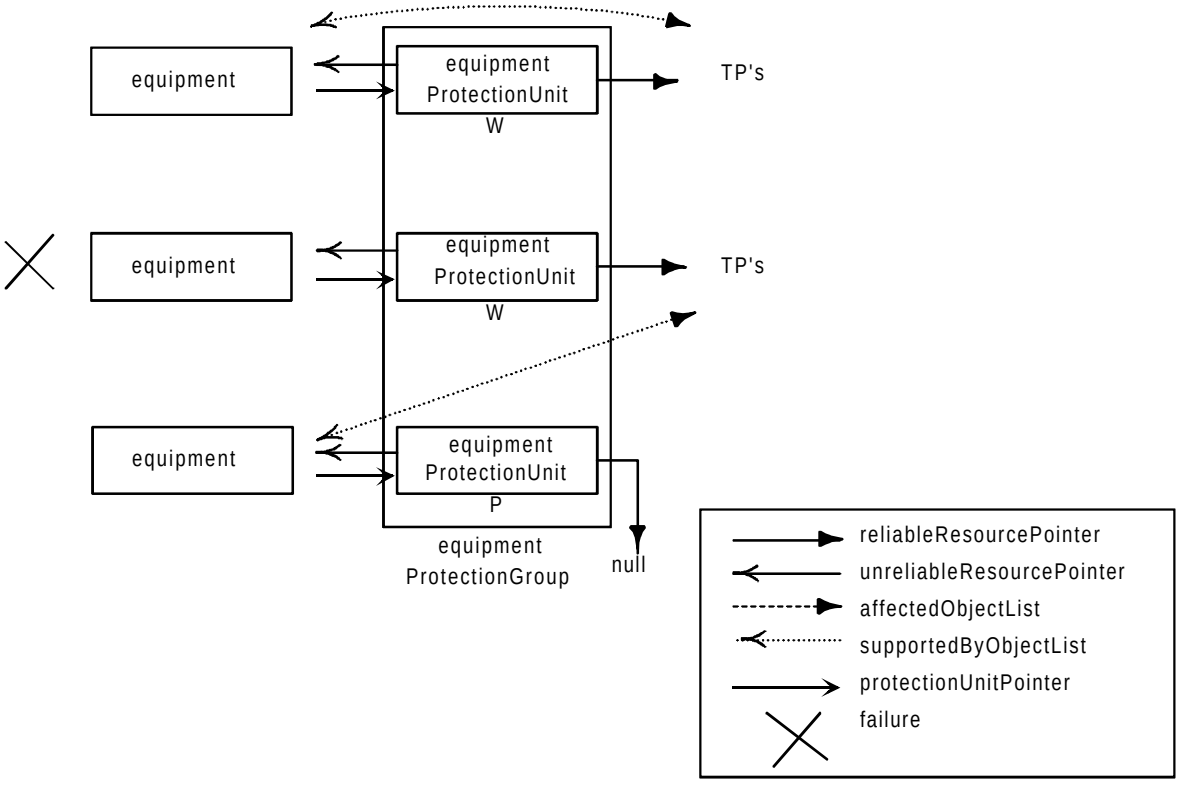


Figure A.14: Equipment protection 1:2 protection

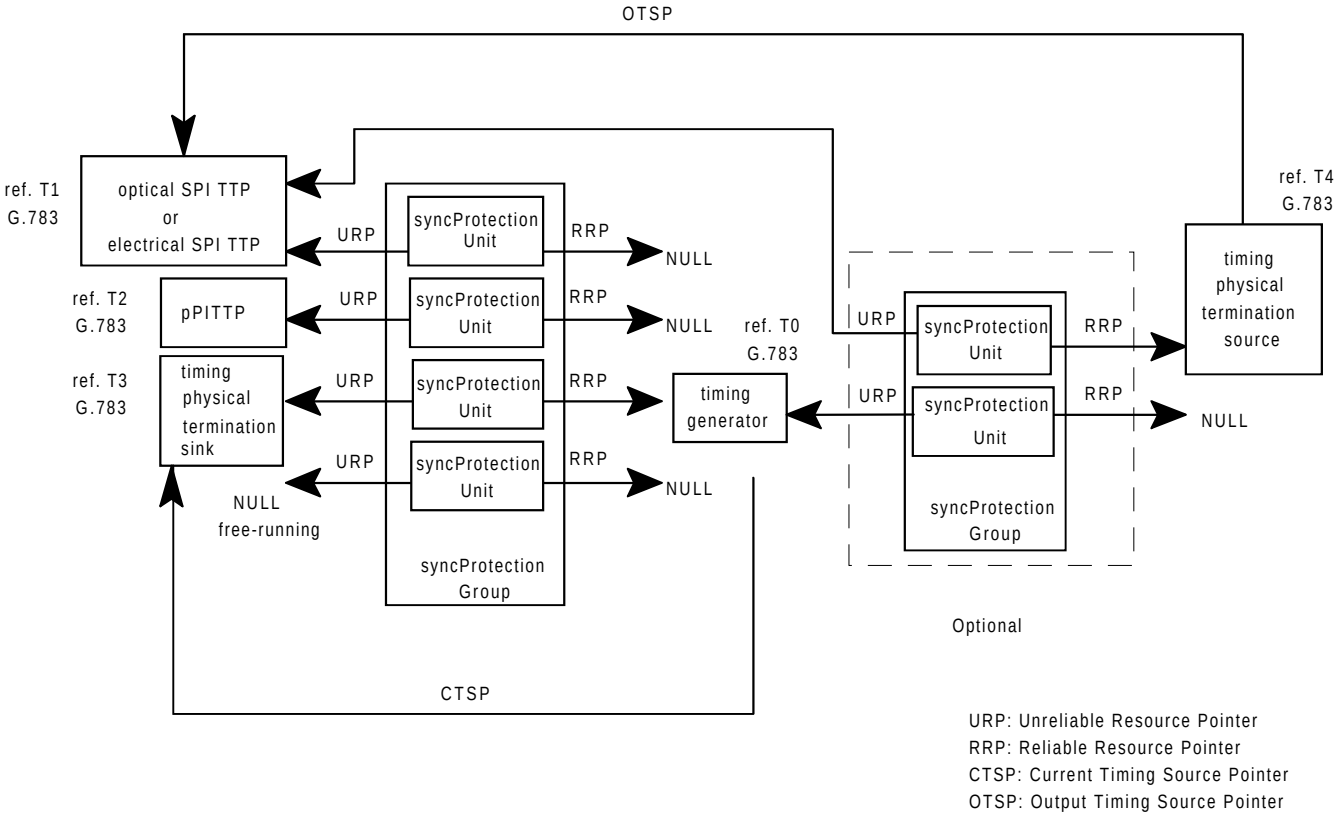
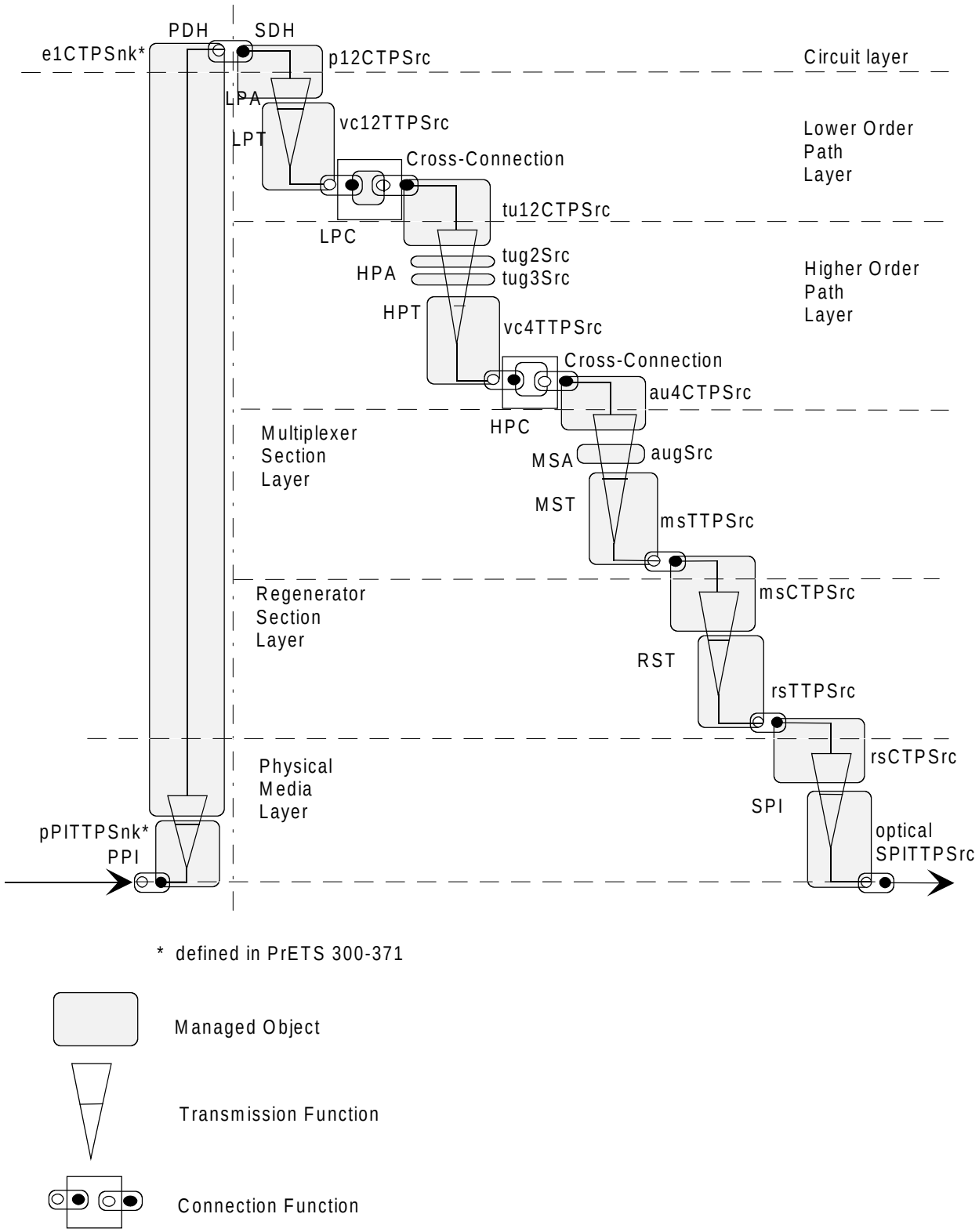
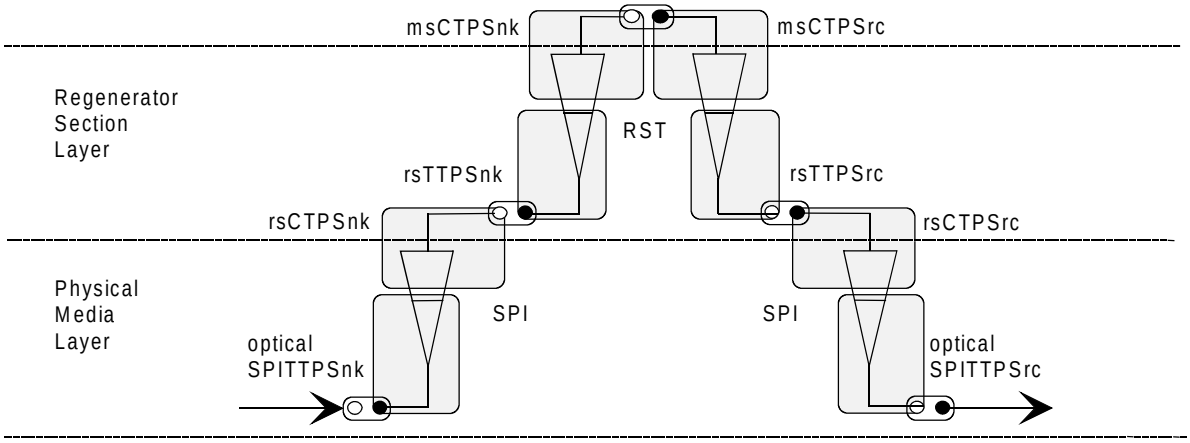


Figure A.15: Timing protection schema in the information model



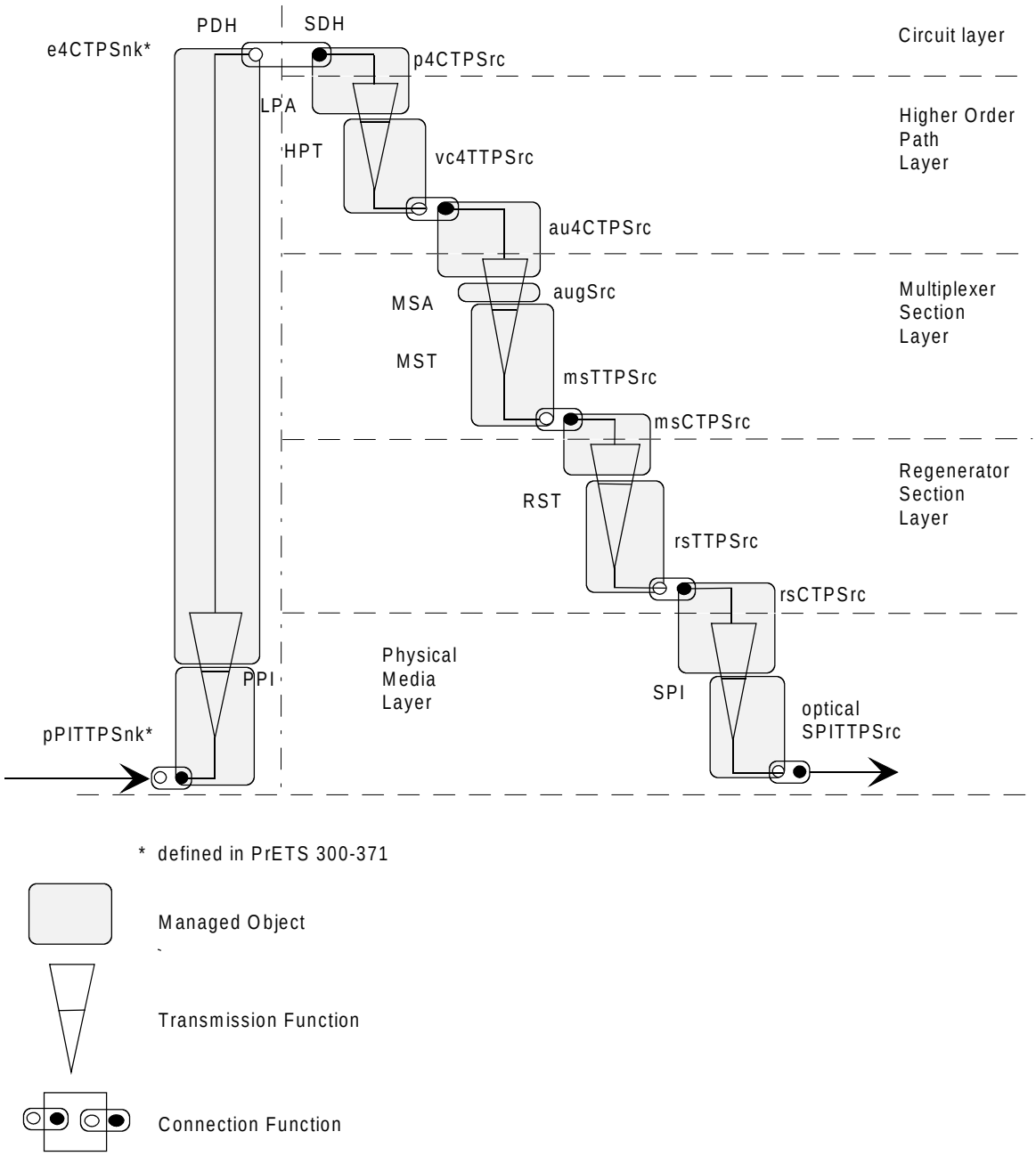
Example 1: 2 Mbit/s signals multiplexed to STM-N signal

Figure A.16: Examples for the relationship between object classes and transmission (continued)



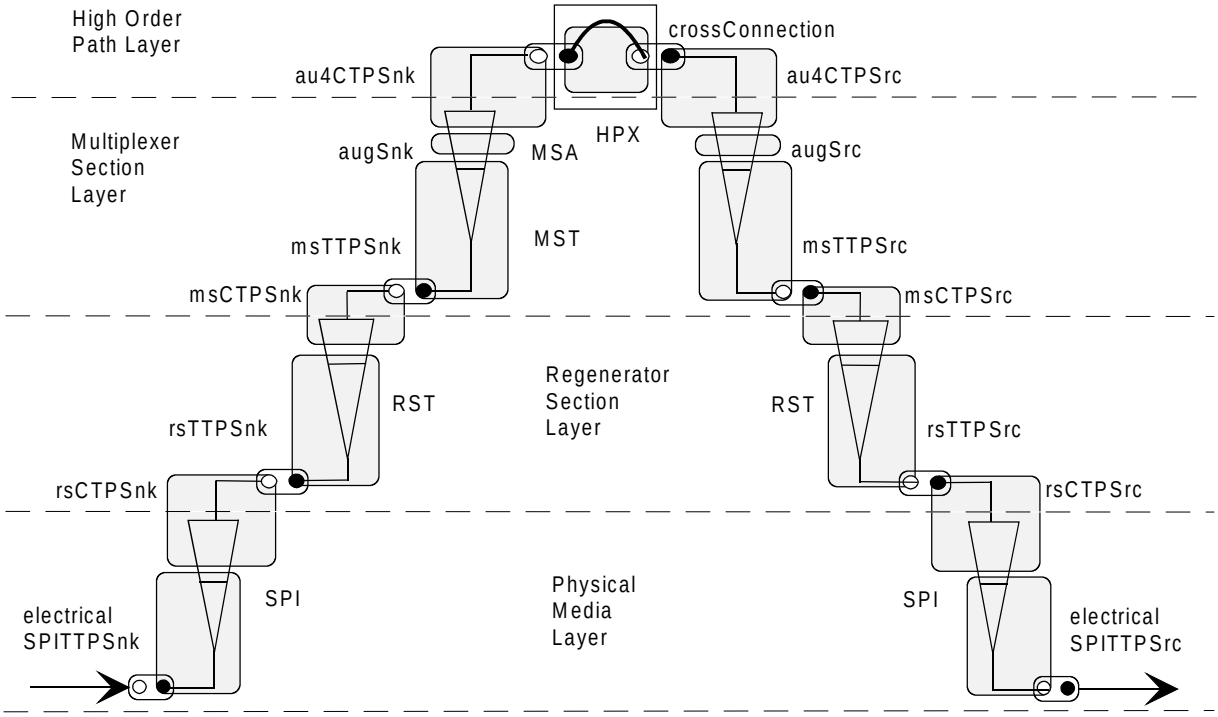
Example 2: STM-N unidirectional repeater

Figure A.16 (continued)



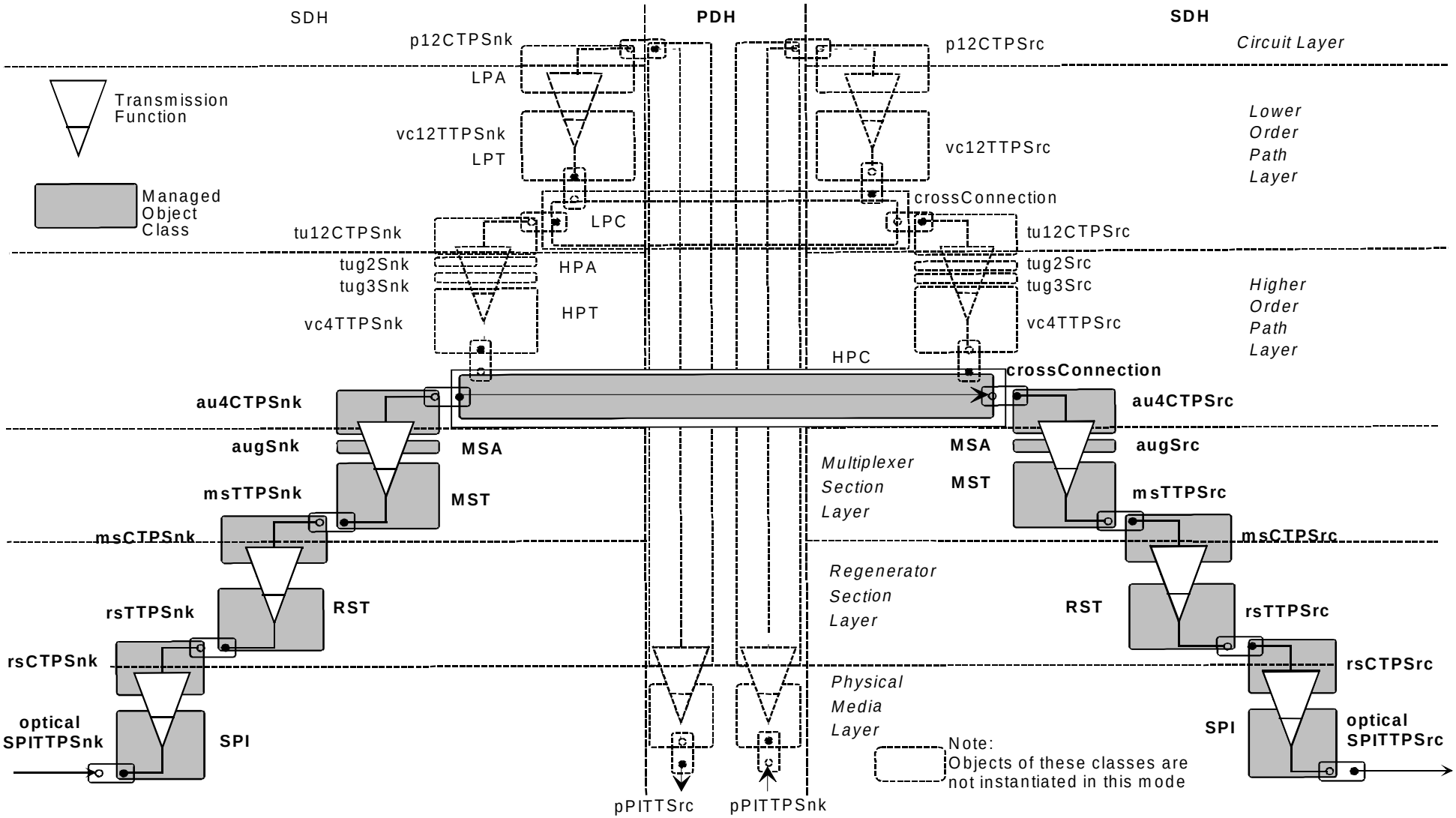
Example 3: 140 Mbit/s signals multiplexed to STM-N signal

Figure A.16 (continued)



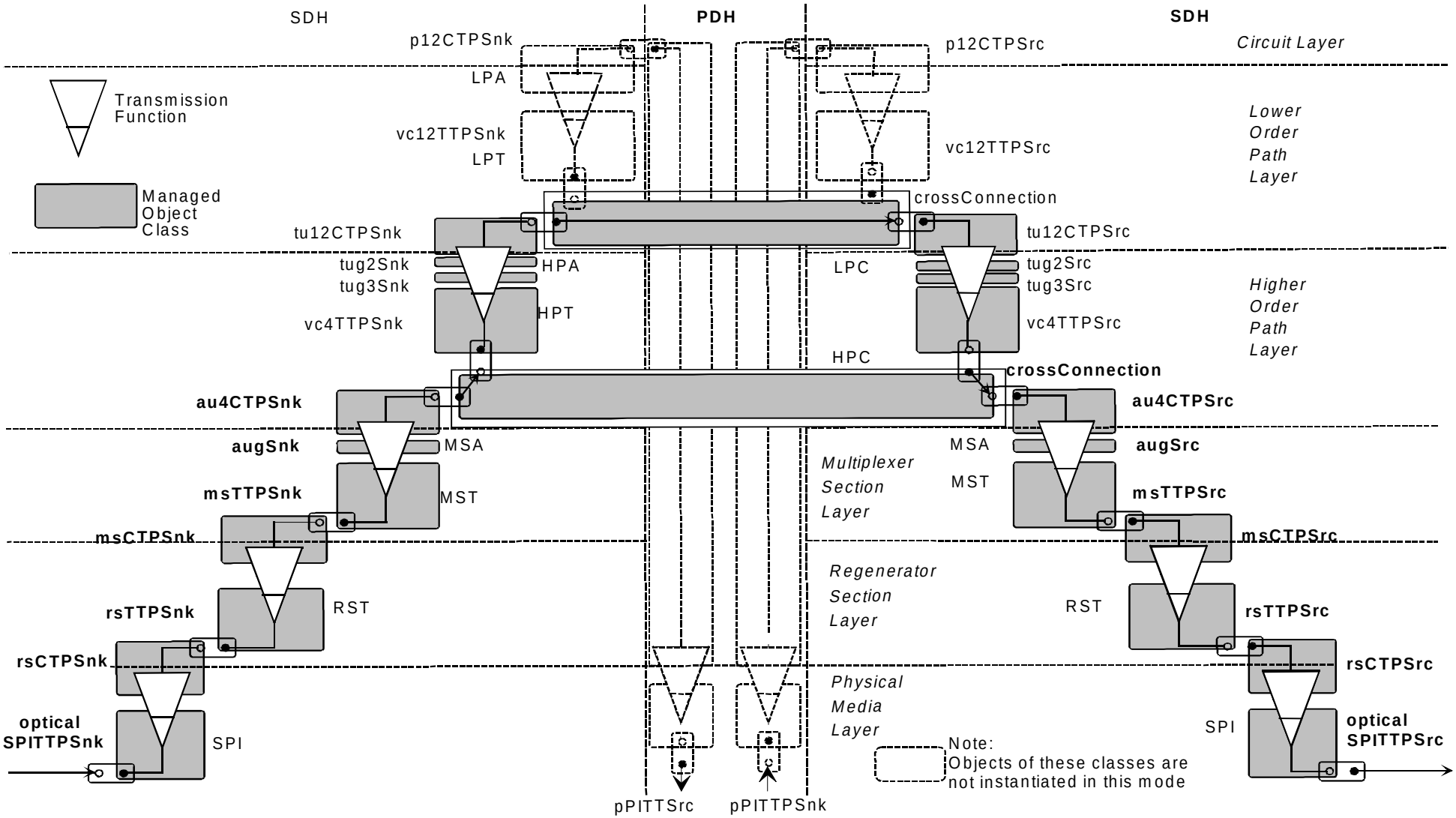
Example 4: SDXC4/4 higher order cross-connect

Figure A.16 (concluded)



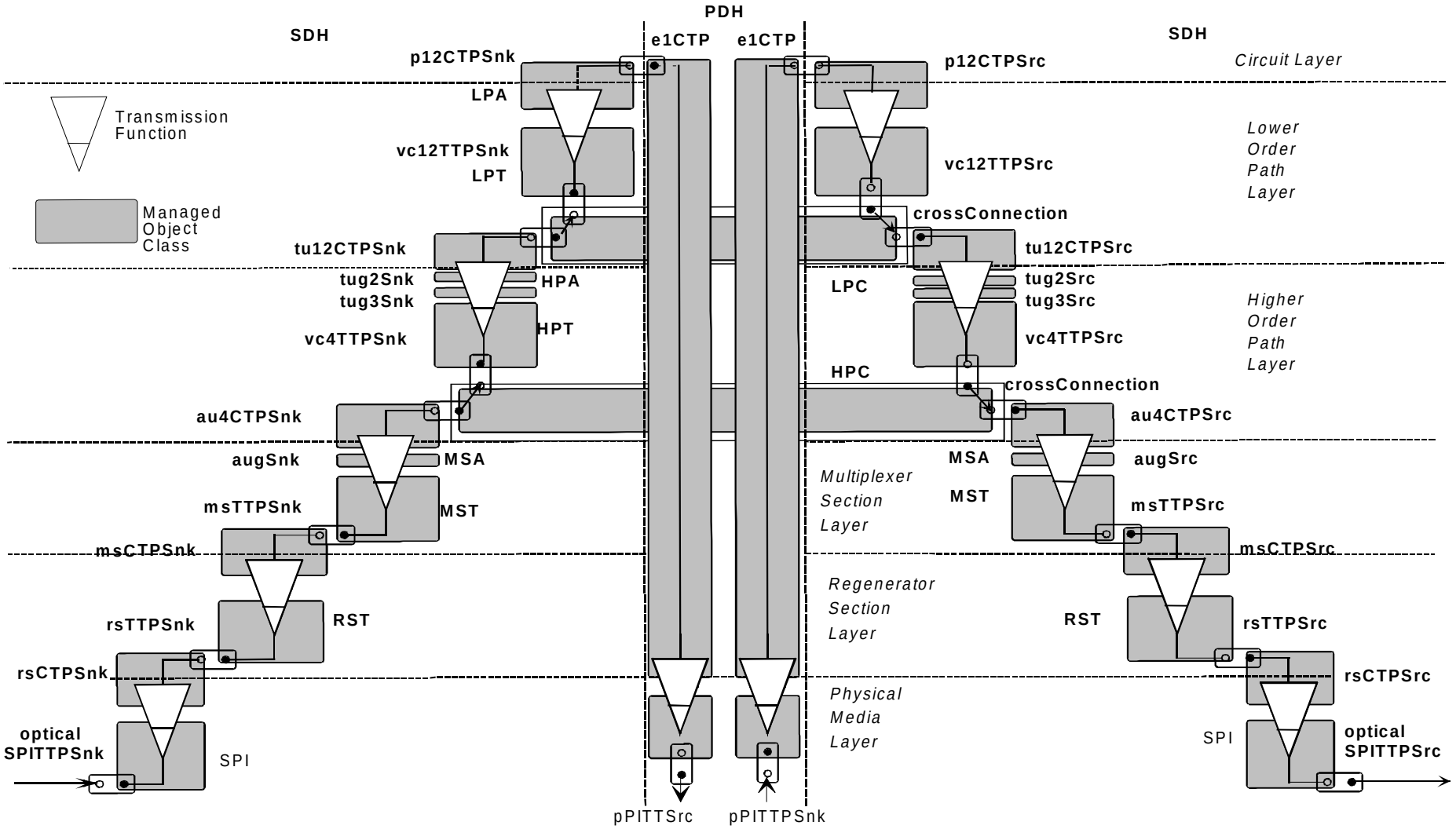
Example 1: Add-Drop-Mux or SDXC4/1 with Au-4 cross-connection active

Figure A.17



Example 2: Add-Drop-Mux or SDXC4/1 with TU-12 cross-connection active

Figure A.17 (continued)



Example 3: Add-Drop-Mux or SDXC4/1 with TU-12 cross-connection active

Figure A.17 (concluded)

Table A.1: Mapping of ITU-T Recommendation G.783 [4] defects on ITU-T Recommendation M.3100 [5] or ITU-T Recommendation X.721 [6] probable causes

Block in G.783 [4]		G.783 [4] defect	G.774 [2] probable cause	M.3100 [5] probable cause
4-1	SPI	Receive loss of signal	LOS	lossOfSignal
4-1	SPI	Transmit fail	-	- Use transmitFailure from X.721 [6]
4-2	RST	Loss of frame	LOF	lossOfFrame
4-3	MST	Multiplex section AIS	AIS	aIS
4-3	MST	Excessive errors	excessive BER	transmissionError
4-3	MST	Signal degrade	signal degrade	degradedSignal
4-3	MST	Far end receive failure	FERF	farEndReceiverFailure
4-5	MSA	Loss of AU pointer	LOP	lossOfPointer
4-5	MSA	AU AIS	AIS	aIS
4-7	HPT	Mismatch of HO path trace ID	path trace mismatch	pathTraceMismatch
4-7	HPT	Mismatch of HO path signal label	signal label mismatch	payloadTypeMismatch
4-7	HPT	HO path FERF	FERF	farEndReceiverFailure
4-8	HPA	Loss of TU pointer	LOP	lossOfPointer
4-8	HPA	TU AIS	AIS	aIS
4-8	HPA	Loss of TU multiframe	loss of TU multiframe	lossOfFrame
4-10	LPT	Mismatch of LO path trace ID (note 2)	path trace mismatch	pathTraceMismatch
4-10	LPT	Mismatch of LO path signal label	signal label mismatch	payloadTypeMismatch
4-10	LPT	LO path FERF	FERF	farEndReceiverFailure
4-11	LPA (note 1)	Frame alignment loss	LOF	lossOfFrame
4-12	PPI (note 1)	Loss of incoming tributary signal	LOS	lossOfSignal
4-14	HPOM (note 2)	Mismatch of HP path trace ID	path trace mismatch	pathTraceMismatch
4-14	HPOM (note 2)	Mismatch of HP path signal label	signal label mismatch	payloadTypeMismatch
4-14	HPOM (note 2)	HO path FERF	FERF	farEndReceiverFailure
4-15	LPOM (note 2)	Mismatch of LO path trace ID	path trace mismatch	pathTraceMismatch
4-15	LPOM (note 2)	Mismatch of LO path signal label	signal label mismatch	payloadTypeMismatch
4-15	LPOM (note 2)	LO path FERF	FERF	farEndReceiverFailure

NOTE 1: Not currently expressed in ITU-T Recommendation G.774 [2].

NOTE 2: Due to revision of ITU-T Recommendation G.709 [1] or ITU-T Recommendation G.783 [4].

Annex B (informative): Bibliography

For the purposes of this ETS, the following informative references have been used:

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- ITU-T Recommendation X.734 (1992): "Information technology - Open Systems Interconnection - System management: Event report management function".
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